Prompt-to-Touch: Towards Enabling Automatic Haptic Effect Generation from Text Prompts Using Text-to-Audio Models

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We introduce Prompt-to-Touch, a proof-of-concept of a multi-step pipeline that generates haptic effects based on the textual description of a haptic experience. Our pipeline first translates the haptic effect description to a sound effect description using our *Foley-Interpreter* component. It then uses a text-to-audio model to generate a sound effect from the sound description. Afterwards, the sound effect is converted into a perceivable haptic effect using our *Dynamic-Audio-Processor*. Finally, the haptic effect is post-processed to compensate for actuator-specific frequency response characteristics. We validate our concept in two preliminary human evaluation studies (n=20, n=10) and a technical analysis. Our results indicate that our pipeline can generate effects to enhance immersive multimedia experiences, abstract desktop/XR interactions, and social communication applications. They also still reveal significant potential for further improvement. Our pipeline could be used to develop future text-driven haptic design and automation tools. We provide open-source code to support future extensions.

CCS Concepts: • **Human-centered computing → Natural language interfaces**; **Haptic devices**; **User interface toolkits**.

Additional Key Words and Phrases: haptic feedback, generative AI, text-to-haptic, large language models, audio-to-haptic, audio models, vibrotactile, extended reality

1 Introduction

Haptic effects can enhance the intuitiveness of mobile and desktop interactions [37, 61, 71, 143], improve the immersion in movie and extended reality experiences [59, 80, 90, 94, 146], as well as convey affective aspects in social communication [44, 86, 116, 138]. They depict brief touch experiences that either augment applications in conjunction with other multi-modal content such as audio and video (e.g. a short rumbling vibration that accompanies a car crash impact in a movie [124]), or convey intuitive metaphorical meaning independent from other feedback modalities (such as a vibration increasing in frequency to indicate the loading progress for a desktop application in the background [4, 5]). However, the creation of haptic effects can be a complex task that

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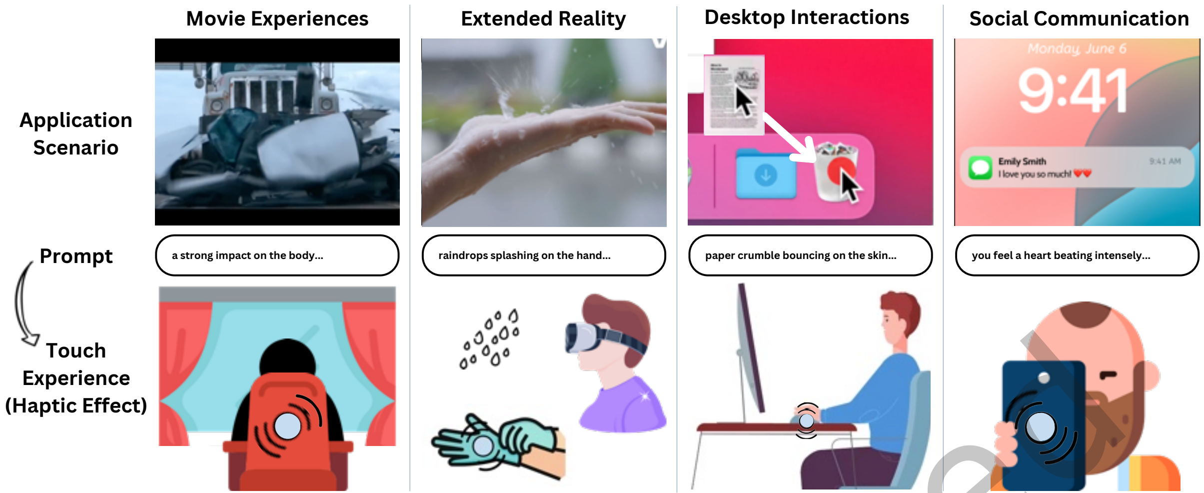

Fig. 1. **Example Application Scenarios:** We provide a proof-of-concept of a multi-step pipeline for generating haptic effects based on natural language text prompts that describe a haptic experience. We show how haptic effects generated using our pipeline can enhance different applications, including immersive movie- and extended reality (XR) experiences, desktop interactions, and social communication.

requires expert knowledge and forms a barrier for haptic feedback design by non domain experts who want to integrate haptics into their end application as well as automation processes [105, 123].

While audio and visual content creation face similar challenges, the recent development in generative text-to-audio [42, 56, 64, 66, 121, 135] and text-to-image models [36, 39, 79, 93, 98, 100, 117, 139] has enabled the development of natural language interfaces that allow for a more simple and natural creation of audio and visual content for various applications. These interfaces are integrated in versatile ways by simplifying the design process for non-domain experts (e.g. a game developer only needs to describe the audio experience to generate a desired sound effect for an in-game event[1]) or fully automating the creation process to augment other media (e.g. enhancing storytelling in text-books by automatically generating images[2])[12, 34, 48].

As for audio and vision, a language interface to generate haptic effects can bring similar benefits for the haptic domain. This includes enabling users to easily personalize and customize haptic feedback in digital systems, simplifying the process of designing new multi-modal experiences for movie producers, game developers, and user interaction designers, as well as providing a platform to easily augment existing audio and video content through fully automated processes. However, recent advances in haptic effect generation have primarily focused on producing effects based on salient features in audio-visual content [53, 62, 140, 141]. The generation of haptic effects based on textual description of a haptic experience yet remains largely unexplored.

In this work we introduce *Prompt-to-Touch*, a proof-of-concept of a multi-step pipeline that can generate haptic effects for a variety of applications solely based on the textual description of a haptic experience (Figure 1). We propose four core components for our pipeline to convert the textual input into a haptic effect that can be played at the skin using a wide-band vibration actuator: First our *Foley-Interpreter* component translates the haptic effect description to a sound effect description. We then use a text-to-audio (TTA) model to generate a sound effect clip based on the sound effect description. After that our *Dynamic Audio Processor* component processes the sound effect clip and produces a haptic effect clip optimized for rendering the haptic experience. In

[1]https://elevenlabs.io/sound-effects
[2]https://www.storybookai.app/

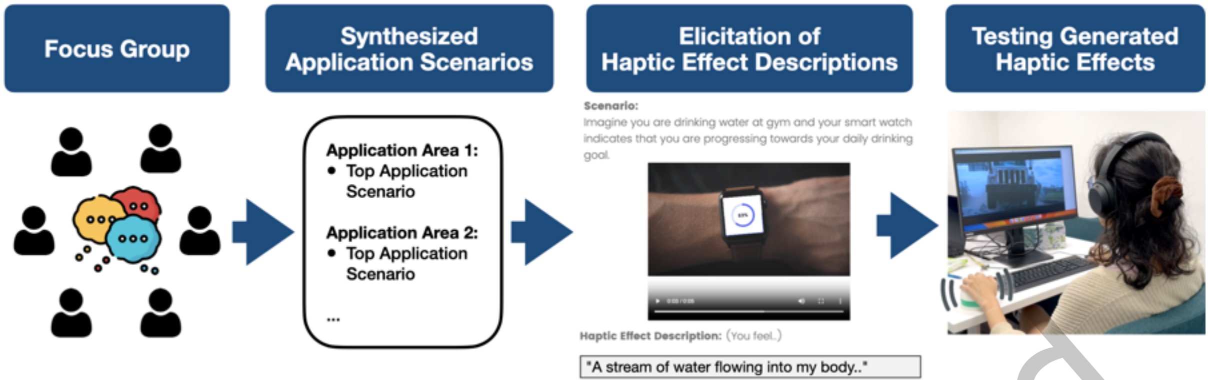

Fig. 2. **Evaluation Method:** We first ideated application scenarios in a focus group with six experts from different domains. We then elicited haptic effect descriptions for the identified scenarios from 10 non haptic experts. Finally we conducted two user studies and a technical analysis to test our pipeline regarding the ability of producing haptic effects to enhance different applications and the performance of individual pipeline components.

a final post-processing step, the haptic effect clip is equalized according to the used vibration actuator to reduce distortion of the haptic effect through actuator specific frequency response characteristics. Our pipeline can form a first building block for future text-driven design and automation tools.

To evaluate our pipeline regarding its capability to generate haptic effects that enhance haptic experiences for different applications, we first collected a test set by ideating applicable application scenarios in a focus group with 6 experts from different domains (Figure 2). Afterward, we elicited haptic effect descriptions for the prior identified scenarios from 10 non haptic domain experts. We created six further scenarios which we believed would demonstrate the effectiveness of our different pipeline components. In two user studies (n=20, n=10) and a technical analysis we evaluate our pipeline based on the collected test set comprising 12 scenarios in 6 different application areas. Our first study shows that our pipeline was able to reflect the haptic effect descriptions and enhance the haptic experience in 10 out of 12 scenarios. We found that our *Foley Interpreter* significantly increased the performance of our pipeline and uncovered potential for improvement of our *Dynamic Audio Processor*. Our second study indicates that our pipeline is able to generate expressive haptic effects by primarily enhancing applications they were specifically generated for (5 out of 6). In conjunction with our second study, our technical analysis provides first insights into the robustness of multiple effect generations, essential for automation processes. While our evaluation indicates that our pipeline depicts a first promising approach to automatically generate haptic effects based on their textual description, it still entails several limitations and opportunities for improvement.

We discuss current limitations of our approach in detail and share insights that could help inform future integrations and extensions. Our pipeline forms a building block that supports further exploration of language based haptic design and automation interfaces. We release the source code of our *Prompt-to-Touch* pipeline[3] to support easy extension and integration. A test page with samples generated using our pipeline is accessible through the web[4]. In summary we contribute:

- A proof-of-concept of a new audio-driven multi-step pipeline that generates haptic effects based on the textual description of a haptic experience.

[3]https://github.com/augmented-human-lab/prompt-to-touch
[4]https://prompt-to-touch.info/test-page

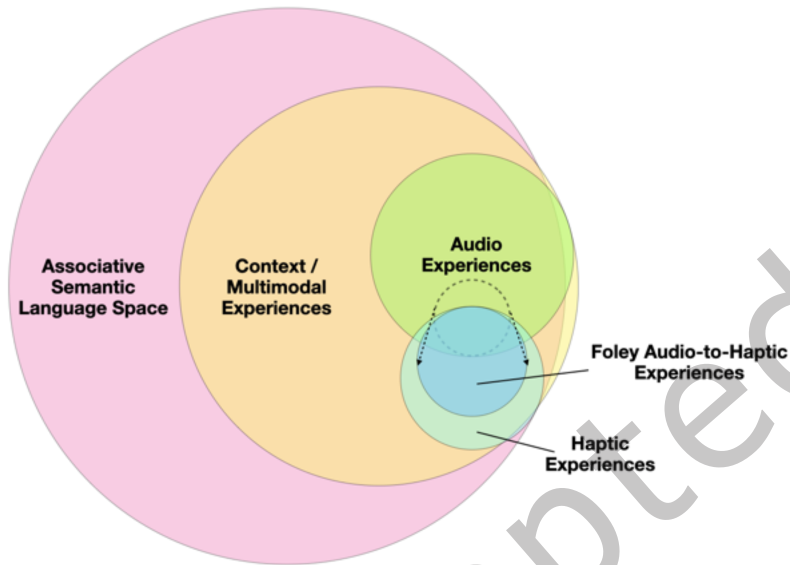

Fig. 3. **Language and the Audio-Haptic Experience Design Space:** (C1) The associative semantic language space allows us to describe (audio and haptic) experiences directly or indirectly through metaphors; (C2) Some haptic effects are not meaningful on their own, but can be meaningful and contribute to a better haptic experience when applied in the appropriate context or harmonizing with other sensory stimuli [96]; (C3) Audio experiences have close parallels with a subset of haptic experiences. Identifying audio experiences that carry characteristics of a haptic experience and applying them in the right context allows projecting them to the haptic experience space by rendering haptic effects effectively using processed sound effects.

- Empirical findings from several user studies and technical assessments with insights on the effectiveness of different pipeline components for generating haptic effects for a variety of application scenarios.

In the following sections, we will first elaborate on the background of haptic effects and its relation to audio effects. After situating our work among contemporary and prior approaches in Section 2, we present the Prompt-to-Touch pipeline in detail in Section 3 by explaining how each component works and how they support the process of generating a haptic effect based on a textual description. After that we describe how we collected an initial application test set (Section 4) and designed our test apparatus (Section 5) to evaluate our pipeline. We then detail our evaluation with participants in two studies regarding the haptic experience (Section 6) and a technical evaluation on the efficiency of the different pipeline components (Section 7). We conclude our work with a discussion on our findings, limitations and potential future directions in Section 8 and Section 9.

2 Background & Related Work

Similar to how sound effects produce an auditory experience through our hearing, haptic effects produce a haptic experience through our sense of touch. Prior work showed that haptic effects can enhance various multimedia experiences and device interactions [37, 44, 59, 146]. The effectiveness of a haptic effect in enhancing an application, however, can highly depend on the applied context and other sensory experiences [52]. For example a haptic effect may not convey meaning independently, however, when harmonizing with other multisensory stimuli and applied in the right context it can be interpreted meaningfully and enhance the overall experience [96]. The most common way to render haptic effects is by using vibration motors. Haptic experiences produced with vibration motors (vibrotactile feedback) share common design patterns and perceptual parallels to audio experiences [4, 5].

Neuroscience research has shown that auditory and tactile processing share neural mechanisms, with integration occurring in auditory cortex [50]. Psychophysical experiments further demonstrate perceptual alignment between the two modalities with regard to frequency perception and harmonics [2, 82]. Hence, certain audio effects played using vibration motors can also produce effective haptic experiences. It is therefore not surprising that several haptic feedback design tools employ techniques akin to those used in sound design to yield haptic effects that leverage the shared audio-haptic design space [24, 90, 124]. Our work bases on the conceptual idea that sound effects often carry characteristics of haptic experiences which has been exploited in several prior work to enrich the haptic experience [9, 84, 140, 141]. While not all haptic experiences can be directly reflected through haptic effects rendered through vibration, even basic effects have shown to be effective when pairing them with other multimodal content [96]. Identifying sound effects that carry similar characteristics as the target haptic experience can therefore enable producing effective haptic experiences, especially within a semantically relevant context (Figure 3). However, manually designing haptic effects directly, or identifying and designing sound effects that suit to enhance haptic experiences indirectly, remains a cumbersome and time-intensive task [96, 105].

2.1 Haptic Design Tools & Libraries

Over the past two decades, a variety of design tools have been developed to simplify the creation of haptic feedback. One of the first design tools, *The Hapticon Editor*, was introduced by Enriquez and MacLean in 2003 [24]. *The Hapticon Editor* allowed to quickly design and replay brief force feedback samples (hapticons) which could communicate a simple idea or emotion. Inspired by auditory icons (earcons, [4]) and sound design software, the user could edit hapticons via a graphical waveform editor. Later work translated these ideas to vibration based feedback and proposed new drag-and-drop interactions as well as UI elements like a tile palette and tile pane to compose haptic icons based on basic templates [99, 123]. With the increasing trend of mobile and spatial interaction, vibration motors have become the most dominant way to provide haptic feedback mostly due to their simple integration, low cost, as well as high size and energy efficiency. While more recent work also increasingly focuses on design toolkits for other types of haptic feedback modalities like forces through pneumatic systems [20, 115], ultrasonic actuators [110, 145], or smart materials [74, 76], most toolkits focus on simplifying the design with vibration motors [49, 68, 85, 89, 127]. Since the vibrotactile feedback modality not only shares similar wave-based characteristics to audio, such as amplitude and frequency [5], but also allows to exploit spatial illusions to create phantom sensation between two actuators, more recent authoring tools increasingly support this too. They reduce the complexity of working with multiple spatially distributed vibration motors and increase the intuitiveness by introducing the concept of direct manipulation of vibrotactile grid arrays [49, 89, 107]. Here, the designer can simply specify properties such as intensity and frequency of the target sensation and the system automatically drives multiple actuators to achieve the target goal. More recent and advanced design tools, like *HapticPilot*, even allow users sketch paths in VR and adapt the feedback to the users current hand pose [122]. Several work explored collaborative toolkits where multiple designers can synchronously or asynchronously design haptic feedback [73, 132, 133]. Schneider et al. explored a realtime haptic design tool which resembles improvising with an instrument [108]. Due to its close relation to and its harmonizing effect with the auditory and visual dimension, haptic feedback is often designed complementary to audio-visual experiences. Several work, hence, focused on design tools on allowing the designer to directly augment audio and visual media content with haptic feedback [18, 125]. While Swindells et al. particularly focused on augmenting dynamic 2D visual media content, Eid et al. proposed an authoring tool, *HAMLAT*, focusing on static 3D objects in virtual environments that can be rendered hapto-visually [18]. With *HAMLAT* the authors furthermore aimed to simplify the design process for non-experts and allow them to focus on specifying haptic properties of an object instead of defining the raw actuation signal itself. Also other work investigated more intuitive haptic design and programming methods. Hong et al. developed a tool relying on demonstration based programming [38, 58]. They argued that opposed to waveform editors which afford accurate design for small samples, demonstration based programming techniques are advantageous when designing long dynamic haptic feedback. Lee et al. proposed the *VibScoreEditor*, which borrows familiar elements from music composition (the piano score and the guitar tablature) to design haptic feedback [60]. With the help of the *Macaron* editor, the authors investigated how designers can leverage examples and simple operations to develop and mix haptic effects [109]. *TECHTILE* comprises an intuitive educational toolkit to easily record and replay tactile feedback [75]. Instead of recording the tactile experience from the surrounding, Degraen et al. proposed allowing the designer to mimic the desired tactile experience directly themselves through vocalization, which then can be modified and converted into a haptic effect [16]. Next to recording the sound as desired effect through vocalization, language, can be a powerful tool to describe the target haptic experience, too.

2.2 Organizing and Describing Haptic Effects Through Language

Describing haptic experiences directly through language is often challenging due to the limitations of everyday vocabulary. For example, conveying the sensation of running a hand across a natural stone involves subtle variations in texture, hardness, and vibration, which are difficult to capture precisely in a few words. To overcome this, both experts and non-experts frequently rely on *indirect descriptions* or *intuitive metaphors*, such as *"feeling the rough surface of a natural stone"* or *"as if a tiny rabbit were hopping"* [7, 29, 45].

Several studies have systematically explored the connection between language and haptic design. Rosenkranz et al. [97] demonstrated that everyday attributes like 'tingling' or 'fading' can be associated with specific vibration parameters, enabling intuitive communication about vibrotactile effects. Similarly, Israr et al. [45] introduced the concept of *feel effects*, pairing linguistic phrases with rendered haptic patterns, showing that non-expert users can reliably generate and recognize a core vocabulary of haptic sensations.

To organize and navigate complex haptic experiences, researchers have proposed structured approaches and visualization tools like *VibViz* and *FeelCraft* [106, 113, 114]. These frameworks integrate multiple dimensions including physical, sensory, emotional, metaphorical and provide designers and end-users with intuitive ways to explore, customize, and author haptic content based on language labels. By combining semantic and parametric spaces, such tools facilitate both the understanding and creation of expressive haptic experiences.

Beyond descriptive vocabularies, data-driven modeling and perceptual profiling allow precise mapping of real or virtual textures to haptic stimuli. The Penn Haptic Texture Toolkit (HaTT) [14] and the haptic texture database by Strese et al. [119] provide recorded textures with associated vibration signals, linking perceptual features to measurable parameters. Similarly, Sakamoto and Watanabe [102] explored tactile perceptual dimensions using vocabulary-guided material sampling, revealing properties such as friction, compliance, and naturalness. Obrist et al. [81] further demonstrated that human experiential vocabulary can be tied to mechanoreceptor responses. Seifi et al. [111] showed that continuous tuning along emotional dimensions (agitation, liveliness, strangeness) can be controlled via vibration parameters, enabling affective haptic design, too.

Rosenkranz et al. extended these concepts to virtual and interactive contexts, demonstrating that haptic feedback in virtual environments can be synthesized to match expected sensory profiles, guided by language-based descriptions [96]. This highlights the potential of natural language as a general interface for authoring haptic experiences, enabling flexible, user-friendly design and interaction across a wide range of applications, analogous to how language is used in audio and visual media.

2.3 Generative AI for Multimedia Content Creation

Recent advances in generative AI have enabled the creation of high-quality content across multiple modalities, driven primarily by large language models (LLMs) and diffusion-based generative architectures. In the visual domain, text-to-image models such as Imagen [100], GLIDE [79], and hierarchical CLIP-based approaches [93] have demonstrated unprecedented photorealism and precise alignment between text prompts and generated imagery. Similarly, text-to-video generation systems, including Imagen Video [36], CogVideo [39], and Make-A-Video [117], extend these capabilities temporally, producing coherent video sequences from textual descriptions, often leveraging pretrained text-to-image models and unsupervised video data to overcome the scarcity of paired text-video datasets. Beyond static and temporal visual content, LLMs have also been used for guiding real-time mixed-reality scene creation [15].

Similarly, Text-to-Audio (TTA) models have recently demonstrated remarkable abilities in generating high-quality sounds that can be controlled using simple text instructions [31, 65, 67]. Traditionally, AI models for sound generation were trained on small, specifically curated music or speech datasets [21–23] that relied on specialized labels for their training procedures. In contrast, TTA model architectures can now leverage larger and more general-purpose audio datasets for training, enabling the generation of a greater variety of sounds. TTA models can be categorized as those that generate music [1, 11, 13, 25, 27] or speech [126], and others that generate general-purpose environmental sound effects such as AudioLDM2 [67] or TANGO [31]. TTA models for sound effects are trained on large YouTube-based weakly captioned audio datasets such as AudioSet [30]. Additionally, to enhance the accuracy of text-based sound effect generation, models undergo training with added text caption augmentations obtained through methods such as crowdsourcing [51], using ChatGPT [70], or utilizing text derived from audio-visual correspondence [10]. Our approach for using AI-generated audio to derive haptic feel effects can use any pre-trained TTA models outlined above.

In this work, we utilize AudioLDM2 [67] to generate meaningful haptic effects, as it demonstrated state-of-the-art (SOTA) performance compared to other existing approaches such as AudioGen [57] or TANGO [31] at the time of this investigation. Furthermore, by leveraging a pre-trained AudioLDM2, we are able to produce a wide range of environmental sound effects, extending beyond just music or speech-based signals.

The idea of bridging text-to-haptic via the audio domain is grounded in how humans naturally connect sensory experiences through language. Writers frequently use metaphors to make descriptions richer and more intuitive, borrowing attributes like "rough" or "bright" from the tactile and visual domains to describe sound characteristics [32, 91]. Key architectural breakthroughs in machine learning, such as attention mechanisms [129] and vector embeddings [88] enable LLMs to link words across sequences and learn abstract semantic relationships, providing the fundament to capture cross-modal associations without experiencing the underlying sensory modalities directly. Prior work has demonstrated that LLMs can translate linguistic metaphors into visual metaphor descriptions that were subsequently rendered by text-to-image models [8], suggesting that similar cross-modal translation may be possible from haptic descriptions to audio. We build on this idea in our *Foley Interpreter* component, recognizing that LLMs cannot yet fully capture the richness of touch from text alone.

2.4 Automated Haptic Feedback Generation

Most prior approaches that automatically generate haptic effects focus on real-time haptic feedback generation based on salient features in audio-visual content. Yun et al. propose a machine-learning classifier that maps vibration patterns and impacts to salient features in the sound signal of video games [141]. Other work focused on rendering realtime spatio-tactile feedback or force feedback based on salient regions in video content to enhance movie and gaming experiences [53, 62, 140]. Also methods to render fine grained tactile textures based on audio-visual features from multimodal content have been demonstrated before [118, 137, 142]. Rosenkranz et al. proposed a first authoring based approach that allows users to quantify well defined design profiles of haptic effects [96]. They demonstrate a bottom up approach to generate vibration patterns by mixing basic vibration features to enhance truck driving simulation in the metaverse. More recent work that arose in parallel to our investigation explores different ways of integrating generative AI in the design of haptic feedback. [26, 78] explore approaches to automatically generate tactile image textures based on image or text that can be used to e.g. stylize 3D models and make them physically tangible by fabricating them using a 3D printer. [131] proposed a system that automatically repurposes real world objects as haptic proxys to provide tangible feedback in mixed reality scenes. Jingu et al. propose *Scene2Hap* which automatically generates vibrotactile signals for objects in a virtual reality scene by analysing the scene and using a hybrid approach to automatically find and generate a vibrotactile sample [47]. Similarly, in *ChatHAP* Lim et al. introduce a conversational chat assistant that supports the designer in automatically constructing and refining haptic effects based on samples in an existing library [63]. With *HapticGen*, Sung et al. leverage an audio-derived dataset and fine-tuned with designer preferences to generate text-driven vibrotactile signals.

In contrast to other contemporary work, we propose a more general and flexible top-down approach with specific focus on a generation pipeline that leverages large text-to-audio models to generate sound effects and translating them to haptic effects through a *Dynamic-Audio-Processor* component. While *HapticGen* focuses on scene descriptions as input, our work expects a prompt describing the haptic experience itself (e.g. through metaphors). To produce a suitable haptic effect, we introduce a two-step *Foley-Interpreter* component that produces an intermediate semantically aligned sound-based representation of the described haptic experience. We contribute a detailed evaluation of various isolated pipeline components focusing on a wide range of application scenarios, revealing different strengths and weaknesses. In the following, we elaborate on the design of our modular and extensible pipeline.

3 Prompt-to-Touch

We introduce a new audio-driven multi-step pipeline that generates haptic effects based on the textual description of a haptic experience. The generated haptic effects can be played using a wide-band vibration motor to render the tactile experience at the skin.

3.1 Overview

To convert haptic effect descriptions to haptic effects effectively, we propose the Prompt-to-Touch pipeline with four core components: (1) *Foley Interpreter*, (2) *Text-to-Audio Generator*, (3) *Dynamic Audio Processor*, (4) *Post-Processor*. Figure 4 provides an overview of our pipeline with the four key components and their functionalities. First, we translate the haptic effect description to a sound effect description using our *Foley Interpreter* component. Then, we translate the sound effect description to a sound effect using our *Text-to-Audio Generator*. Afterward, we convert the sound effect to a perceivable haptic effect using the *Dynamic Audio Processor*. In the final step, we remove frequencies beyond our touch perception and post-process the haptic effect so they can be played on a target actuator using our *Post-Processor* component. We describe each component in detail in the following.

3.2 Foley Interpreter

The first component of our pipeline is the *Foley Interpreter*. Inspired by Foley in filmmaking, a post-production technique to recreate and match sounds to movie scenes that provide a better audible experience than the original sound effect itself, our *Foley Interpreter* converts a simple haptic effect description to a sound effect description.

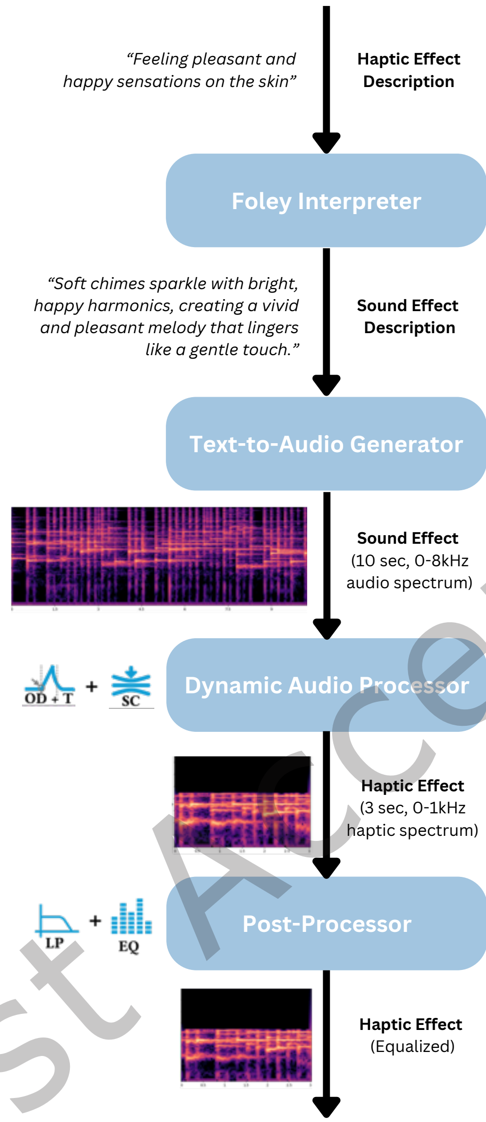

Fig. 4. **Prompt-to-Touch Pipeline Schematic:** It comprises four core components including *Foley Interpreter*, *Text-to-Audio Generator*, *Dynamic Audio Processor*, and *Post-Processor*. Spectrograms show the frequency spectrum (log-scaled y-axis in Hz) over time (x-axis in seconds) for the generated waveform after the individual generation and processing steps.

The goal of the *Foley Interpreter* is to produce a sound effect description that can be fed to the *Text-to-Audio Generator* component to generate a sound effect that is meaningful when rendering it as a touch experience.

The *Foley Interpreter* component is implemented using *GPT-4o*, a large language model (LLM) which has been trained on vast amounts of data from the internet that allows it to understand and reason in various domains. As discussed in Section 2, the shared semantic space between haptic and auditory descriptions motivates the use of LLMs for this translation task. While LLMs do not directly experience touch or sound, they can leverage cross-modal associations learned from human text to translate haptic descriptions into semantically aligned sound effect descriptions. To elicit a textual response from *GPT-4o* we utilize the OpenAI Python API. We first specify a *System Message* to define the role and set the focus of the LLM to our target domain (haptic feedback). Subsequently, we provide an *Instruction* for the task. During initial tests we noticed that the LLM is prone to focus on the wrong aspects of the haptic effect within a single step. We therefore split the task for the LLM into two steps. In the first step, the *Foley Interpreter* breaks down the target haptic experience into its core components and outlines potential sequential elements that can be played in audio step-by-step. We leverage chain-of-thought (CoT) a prompt engineering technique that has been shown to improve reasoning quality in LLM responses. The used prompt including system message and instruction can be found in Appendix A (Step 1).

In the second generation step, the LLM is fed the result of the first step and instructed to identify the most characteristic aspect of the target experience and formulate an audio description that focuses on vocabulary of audio captions. By key characteristics we refer to the spectral and intensity patterns that evolve over time and define the perceptual quality of the expected audio-haptic experience. When the haptic experience involves a sequence of events, the *Foley Interpreter* is additionally intended to identify the most characteristic step of that sequence, i.e., the step most likely to evoke a strong perceptual association with the intended experience. For instance, when the target is to create the haptic experience of feeling a scratchy wall, the *Foley Interpreter* may describe a sequence consisting of (1) the finger first colliding with the wall, then (2) the fingers moving along the scratchy surface. Here, the most characteristic aspect people associate with this experience is likely the second part, where they feel the wall texture, and not the initial touch-down event. The *Foley Interpreter* is then intended to characterize the perceptual qualities of this step, such as the irregular higher-frequency vibration patterns evoking roughness, and translate these into a corresponding audio description. Within initial testing we found that without these steps the pipeline may create haptic effects highlighting parts of the experience that result in less powerful or even wrong associations. After completing step 2, we derive audio captions as a result, which serve as the primary training data format for the TTA model we employ in our next pipeline step. The used prompt including system message and instruction for our *Foley Interpreter Component* can be found in Appendix A (Step 2).

During testing we noticed that the LLM produces very long audio descriptions for short haptic effect descriptions that contain many irrelevant details. While we explicitly instruct the LLM to produce shorter descriptions, it still produces relatively longer descriptions. However it tends to focus on more relevant details. This was to our satisfaction as recent work shows that longer audio descriptions can have a positive impact on the output quality of TTA models[5].

Based on this implementation our *Foley Interpreter* interprets a haptic effect description *"Feeling pleasant and happy sensations on the skin"* as the sound effect description *"Soft chimes sparkle with bright, happy harmonics, creating a vivid and pleasant melody that lingers like a gentle touch."* (see Figure 4). While the former describes an inaudible experience, the latter uses sound elements like instruments to translate the key characteristics into an audible experience.

3.3 Text-to-Audio Generator

In the second step of our pipeline, we use a text-to-audio (TTA) generator. This generator takes in the sound effect description and produces a contextually relevant audio sample. For this purpose, we employ a latent diffusion-based [95] TTA model called *AudioLDM2* [67] which generates a 10-second audio sample between 0-8000Hz with a sampling rate of 16kHz. Diffusion-based TTA models are probabilistic in nature. They produce a different sound sample each time, even when given the same sound effect description. To ensure that the

[5]Most TTA model demos encourage using long descriptive sentences to nudge the models to generate better sounds. E.g. Tango: https://huggingface.co/spaces/declare-lab/tango.

generated sound faithfully incorporates the semantic concepts expressed in the sound effect description, we run the generation process three times and select the sound with the highest CLAP (Contrastive Language Audio Pre-training) [134] score. Additionally, each generation for each sample goes through 100 de-noising diffusion steps. At the end of this pipeline step we hence obtain an audio sample from the sound effect description that embeds the key characteristics of the described haptic effect. However, the characteristics can be embedded in the audible range above 1000Hz which is the upper threshold for our tactile perception Additionally, the 10-second samples are generally too long for shorter haptic effects and the key characteristic may be contained anywhere within the 10-second sound. Hence, we incorporated a Dynamic Audio Processor step that addresses these aspects.

3.4 Dynamic Audio Processor

This step of our pipeline processes the generated audio sample through a series of signal-processing functions to generate a haptic effect sample. Prior work has investigated methods to translate audio from the audible range of 20Hz to 20kHz to the haptically perceivable range of 0Hz to 1000Hz using frequency shifting [72, 83]. However, static frequency shifting moves all frequencies by a constant amount and may not guarantee that key signal characteristics end up in the tactile range. Okazaki et al. report that frequency shifting works for sounds with high-frequency components but produces unwanted distortion for low-frequency dominant sounds [83]. Since our generated sounds may be dominant in high frequencies, low frequencies, or both, we opted for a dynamic spectrogram compression method that can shift high-frequency components down while retaining low-frequency components.

Onset Detection (OD) and Trimming (T): As the generated audio sample is 10 seconds long, and haptic effect responses often last a few seconds, we trim our audio samples down to three seconds. For this, we first reduce the background noise in the 10-second AI-generated audio using noise gating functions [101]. We then detect the first sound event in the denoised sound and clip three seconds from it to generate the haptic effect. For instance, for the sound of *"Slow notes played by a melancholic piano"*, we select the first three seconds of the audio starting from the first onset of a musical note in the sample. The denoising process helps focus on the prominent events (here notes) while ignoring the noisy background events during trimming.

Spectrogram Compression (SC): Environmental and musical sounds, such as sounds of wind or chimes, are usually associated with high-frequency components (typically above 2-3kHz). To convert these sounds to perceivable haptic effects, we employ 'average pooling compression' on the 2D spectrogram image representation of the sound. This reduces the spatial dimensions of the spectrogram image while preserving important semantic aspects of the objects in the image, such as object contours and their relative position. This way we shift sound events in the higher frequencies to a lower frequency range.

Our implementation first converts the spectrogram to raw audio waveforms using the spectrogram inversion method [33] and then compresses frequency components from zero to the mean centroid of the spectrogram to the $0 - 1000$ Hz range.

Compared to other methods like frequency shifting [84], the average pooling technique allows us to preserve information of the lower frequency spectrum and not clip off events that could be meaningful for rendering the haptic experience.

At the end of this pipeline step, we hence obtain a processed haptic effect sample that is 3 seconds long, compressed to the haptic perceivable range, and should contain the key characteristic of the described haptic effect. However, most vibration actuators have a resonant frequency which can distort the rendered audio sample and hide important characteristics of the intended haptic effect. We hence employ a final post-processing step which aims to compensate for that.

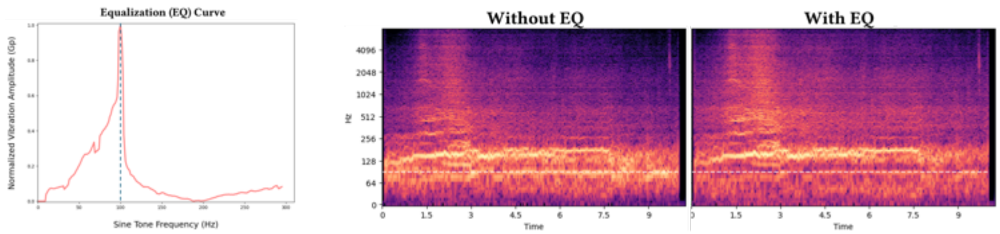

Fig. 5. **Equalization Curve:** The equalization curve for our actuator and the effect of equalizing a 10-second sound sample, based on the curve, for the prompt "A car crashes into another car". The gain in the spectrogram with EQ is reduced along the 100Hz mark, where the vibration amplitude is maximum in the EQ curve.

3.5 Post-Processor

The last component of our pipeline is the post-processor. Note that this stage is not integral part to our proposed pipeline, but rather a step to compensate potential distortion through the frequency response characteristics of the used vibration actuator (equalization step) and to remove audible aspects above 1000Hz (low pass filter step).

This step may not be required if the actuation system already compensates for this, similar to what many audio systems do. While the Dynamic Audio Processor already compresses sounds to 0-1000Hz, we still apply this step to any ablations that may not include our *Dynamic Audio Processor* component to ensure that most audible aspects above 1000Hz are removed.

Equalization (EQ): To equalize generated sound samples, we recorded the frequency response characteristics of the employed actuator using an MPU-9250. Based on this data, our *Post-Processor* equalizes all sounds generated by the TTA model; that is, we reduce the gain (or loudness) of the sound for the band where actuators vibrate more strongly. With this equalization, we ensure that sound events in all frequency bands are more equally represented within the haptic response. Figure 5 shows the frequency response of the actuator we used for our haptic test apparatus. We display the spectrogram representations of the sound for the text prompt "A car crashes into another car" with and without equalization. The vertical dotted line (around 100 Hz) shows the peak of the curve. Similarly, the dotted white line across the spectrogram represents 100 Hz across all time steps. Note how, near the peak of the equalization curve around 100 Hz, where the actuator vibration amplitude is highest, the gain is reduced in the spectrogram.

Low Pass Filter (LP): For the low pass filter we limit our sound events within the $0 - 1000$ Hz frequency range as suggested by [84]. We use the scipy library [130] implementation of the Butterworth lowpass filter with a higher order of N=17 for a steeper spectral roll-off.

4 Application Scenario Test Set

In order to get an initial test set and evaluate our proposed pipeline, we collected a total of 12 scenarios covering six application areas with two scenarios each (Table 1). To allow to better compare across different scenarios, we chose for a within subjects design where each participant performs all conditions. This offered to minimize the effect of subjective differences between individuals. To balance between study time and the number of trials per participant, we tried to identify a single haptic description that is most representative for each scenario (as described below), instead of testing all collected haptic descriptions. Initially, six different application scenarios across various areas were identified through a focus group with non-haptic experts who were interested in applying haptics. Subsequently, we gathered haptic effect descriptions for each of the identified scenarios. In addition to the six focus group scenarios, we (as haptic experts) independently generated six further application scenarios inspired by prior work, including their haptic effect description assuming they would demonstrate the effectiveness of our different pipeline components. For each individual scenario we crafted videos to establish a meaningful multisensory context which is important for interpreting the haptic experience as elaborated in Section 2.

4.1 Focus Group: Ideating Applications

We recruited a total of six focus group participants (2 male, 4 female) aged $M = 30.67$ ($SD = 6.92$) with a variety of backgrounds and interests as described in Table 2. What they had in common was that they shared an interest in applying haptics in their respective area, some having experience in applying haptics before.

The focus group followed the nominal group technique [3, 128], a validated method for systematically generating a representative set through group consensus, and was structured as follows: First the focus group conductor welcomed the participants and explained the goal of the focus group being identifying meaningful diverse applications for a system that can produce haptic effects based on textual descriptions of the target haptic experience. After a short introduction, the participants were tasked to first start ideating potential applications individually for three minutes. They were asked to try come up with at least three different application scenarios and think about what application area the scenario would fit in. After three minutes each participant presented their scenarios without interruption, followed by a short group discussion. The individual ideation and discussion round was repeated one more time. All application scenarios and application areas were collected and clustered. Using a prepared survey form participants were then asked to (1) rank the application scenarios of each application area cluster based on their relevance for the application area, (2) rank the application area clusters based on how useful it would be to be able to automatically generate haptic effects for this application area using the proposed system.

Overall, seven clusters with 32 unique application scenarios were identified. The 7 clusters were: *Immersive Experiences in Movie/Gaming/XR*, *Health & Wellbeing*, *Education & Training*, *Augmenting Digital Interactions*, *Social & Empathetic Communication*, *Music & Art, Workplace & Industry* (ranked order). Since all *Education & Training* scenarios were based on spatial guidance we excluded this application area. For example the top scenario related to education and training envisioned guiding someone to play violin with the correct finger using spatial feedback at the respective finger. However, as our initial investigation aimed to focus on a single actuator this was inapplicable. This also provided us with insights in the limitations of our current approach which we discuss at the end of this work. After eliminating *Education & Training* scenarios, the final application set comprised six application scenarios (one for each remaining application area) generated by the focus group (Table 1). Corresponding visuals for the application scenarios are displayed in Appendix B.

4.2 Eliciting Realistic Haptic Effect Descriptions

To elicit realistic haptic effect descriptions, we asked 10 non haptic experts (4 male, aged $M = 29.30$ $SD = 3.89$) to try describe the target haptic experience for the previously identified scenarios. All but one reported prior experience with haptic feedback; mostly through smart phones or gaming controllers.

To provide a meaningful multisensory context (section 2), we first carefully crafted short 3-5 second videos that illustrate the six application scenarios from our preceding focus group. For each of the scenarios, participants were provided with a description of the scenario, and a video to help them visualize the scenario in a Qualtrics survey. They were then given the task to "Describe a suitable haptic effect on the skin that is unique, intuitive to understand, and fits this specific scenario.". The participants were made aware that they could describe the haptic effect directly or use associations and metaphors to describe the haptic effect indirectly.

In total we collected 60 haptic effect descriptions, ten for each of the six application scenarios. From the 60 collected descriptions we filtered out haptic descriptions that relied on mapping haptic events to distinct audio-visual features of the scenarios or were solely based on a spatial rendering method. After filtering we

#	Application Area	Scenario Description	Haptic Effect Description	Creator
1	Immersive Experiences in Movies / Gaming / XR	"Imagine you are in a virtual reality game and touching the soft fur of a cat."	"The soft fur softly rubbing against the skin corresponding to the cat's fur"	Focus Group
2	Health & Wellbeing	"Imagine you are drinking water at gym and your smart watch indicates that you are progressing towards your daily drinking goal."	"A slow vibration slowly increasing until it reaches the current level of drinking water"	Focus Group
3	Augmenting Digital Interactions	"Imagine you are in a extended reality environment and scroll through a menu by rotating a virtual knob."	"similar short pulses of medium intensity to corresponding to the clicks"	Focus Group
4	Social & Empathic Communication	"Imagine you are at home and you receive a message from your significant other on your phone."	"Soft vibrations felt on the skin according to a heart rhythm"	Focus Group
5	Music & Art	"Imagine you are practicing piano and using a metronome to help you staying in the rhythm. An assistive device renders the rhythm of the metronome as touch sensations on your body."	"rhythmic tap on the skin"	Focus Group
6	Workplace & Industry	"Imagine you are working in a factory. A machine fault occurs. In addition to a visual red light at the machine, you are wearing a device that renders a touch sensation as warning to you."	"long and strong pulses"	Focus Group
7	Immersive Experiences in Movies / Gaming / XR	"Imagine you are in a virtual reality game and you are outside in the rain."	"Raindrops falling on the hand."	Authors [35, 43, 87]
8	Augmenting Digital Interactions	"Imagine you are sitting in front of your computer and drag a desktop file into the virtual trashcan to delete it."	"You feel a paper bouncing on your skin like it is thrown into a trashcan."	Authors [4, 5]
9	Immersive Experiences in Movies / Gaming / XR	"Imagine you are in a virtual reality game and touching a rock."	"feel light scratches along a surface"	Authors [46, 120]
10	Immersive Experiences in Movies / Gaming / XR	"Imagine you are watching a movie and seeing a bird pecking a surface."	"feel a bird pecking lightly on the skin"	Authors [45, 77]
11	Social & Empathic Communication	"Imagine you receive a happy message on your phone and your phone notifies you about it."	"feeling pleasant and happy sensations on the skin vividly"	Authors [19, 111]
12	Augmenting Digital Interactions	"Imagine you are working on a desktop computer and sending off an email. The computer notifies you that the email has been sent off."	"you feel a short thrust burst of a rocket launch fading away. like a swoosh"	Authors [6, 24]

Table 1. **Application Scenario Test Set:** Our test set consisting of 12 application scenarios with their respective scenario and haptic effect description. Visuals for each application scenario are displayed in Appendix B.

ID	Background & Interests
P1	A researcher with a background in psychology who is interested in applying haptics in health and education.
P2	An embedded engineer and product designer who has experience working on multiple commercial projects involving haptic feedback.
P3	A human-computer interaction researcher with focus on biofeedback and emotions pertaining to wellbeing, having an interest in applying haptic feedback in one of her future projects.
P4	A doctoral student with a background in engineering, having an active interest in integrating haptics with focus on accessibility and robotics applications.
P5	An Extended Reality (XR) specialist with over five years of experience in designing interactions and immersive experiences in XR including applying haptic feedback in multiple projects.
P6	A doctoral student with a broad perspective on cognition, emotions and artificial intelligence, interested in understanding how haptic feedback can influence the human mind and emotions.

Table 2. **Focus Group Participants:** Background and interests of the six focus group participants to ideate application scenarios.

Fig. 6. **Haptic Test Apparatus:** We used a Drake LF Wideband vibration motor ($10Hz$ to $> 300Hz$) embedded in a 3D printed haptic dome with a silicon layer to play generated haptic effects to the inner hand palm and fingers of participants.

were left with at least six haptic descriptions for each scenario. To shrink down the set and be able to run a user study, we identified a single most representative haptic effect description for each scenario by (1) creating vector embeddings using (llama3) for each description, (2) getting the sum of the pairwise similarity scores for each description using FAISS, and (3) choosing the description that is most similar to all other descriptions.

This allowed us to yield a single most representative haptic effect description for each scenario that was generated by non experts and that we could use to generate haptic effects using our pipeline. Including test scenarios with their respective description by the authors, we ended up with a total set of 12 application scenarios that we wanted to test. All scenarios with haptic effect descriptions can be found in Table 1.

5 Haptic Test Apparatus

To test haptic effects generated using our pipeline with users, we used a Low Frequency (LF) DRAKE Wideband vibration motor and the TITAN Core Development Kit. The kit allows to play audio waveforms on the actuator wireless via an ESP32. Based on pilot testing we found that the LF actuator could render different kind of haptic effects best, including low frequency effects (such as impacts 0 to 100Hz) as well as higher frequency effects (100 to >300Hz, carrying more high fidelity characteristics from e.g. raindrop sensations and harmonics of instruments). Since the actuator has an inherent peak resonant frequency between 90Hz - 100Hz which can distort the raw waveform signal, we recorded the response frequency spectrum of the actuator using an MPU-9250 and equalized all our haptic effects as described in the post-processor implementation in Section 3.5.

To validate our pipeline across different scenarios in a consistent way, we integrated the DRAKE vibration motor into a 3D printed haptic dome (a hemispherical shell, see Figure 6). By placing the hand on the haptic dome users can experience the haptic effect through their hand and fingers. A silicon layer provides an elastic element allowing the haptic dome to resonate with the actuator even if pressure is exerted through the hand touching the dome. The latter would otherwise restrict the movement of the dome. We share the printable 3d model to easily reproduce our apparatus[6].

6 Human Evaluation

6.1 Study 1: Enhancing Haptic Experiences & Validating Pipeline Components

The objective of our first study was to investigate the ability of our pipeline to generate haptic effects based on haptic effect descriptions and to what degree generated effects can enhance the haptic experience across different applications. In addition, we wanted to assess how individual pipeline components, specifically the *Foley Interpreter*, the *Text-to-Audio Generator*, and the *Dynamic Audio Processor* influence the experience of the generated effects. To do so we wanted to collect participant ratings about the haptic experience in context of the audio-visual scenarios, as well as how well the generated effects reflect their textual description. This study was approved by the institutional review board at the National University of Singapore (NUS-IRB-2024-570).

6.1.1 Design. We conducted a within-subjects study design with 12 SCENARIOS × 5 HAPTIC_EFFECT_TYPES × 20 PARTICIPANTS = 1200 samples in total, 60 trials per participant. For each of the 12 scenarios of our application scenario test set, we generated five haptic effect types based on the elicited haptic effect descriptions (section 4). These included four ablations (different combinations of our pipeline components) that always contained the *Text-to-Audio Generator*, were trimmed (first stage of *Dynamic Audio Processor*), and post-processed using our *Post-Processor* for equal comparison of samples. The ablations were as follows:

- **basic_none**: Haptic effects are generated based on the haptic effect descriptions directly. (*Text-to-Audio Generator*, *Post-Processor*)
- **basic_compression**: Haptic effects are generated based on the haptic effect descriptions directly and applying the *Dynamic Audio Processor* compression step. (*Text-to-Audio Generator*, *Dynamic Audio Processor*, *Post-Processor*)
- **foley_none**: Haptic effects are generated based on the audio effect descriptions produced by the *Foley Interpreter*. (*Foley Interpreter*, *Text-to-Audio Generator*, *Post-Processor*)
- **foley_compression**: Haptic effects are generated based on the audio effect descriptions produced by the *Foley Interpreter* and applying the *Dynamic Audio Processor* compression step. (*Foley Interpreter*, *Text-to-Audio Generator*, *Dynamic Audio Processor*, *Post-Processor*)

We also included one baseline test sample with a constant vibration at $250Hz$ at 50% amplitude level.

Measured was the HAPTIC_EXPERIENCE based on the five experiential dimensions (hedonic factors that influence experience) according to Kim et al., 2020 [52], as well as the DESCRIPTION_MATCHING on a 7-point scale (1 worst,

[6]https://www.thingiverse.com/thing:7159263

7 best). Trials were fully randomized. Participants took a break every 15 trials. Actuator noise was shielded (see procedure). The study apparatus was kept consistent across all application scenarios.

6.1.2 Participants. We recruited 20 participants aged $M = 29.55$ years ($SD = 5.24$). 10 female, 10 male. Two were left-handed. 14 reported basic experience with haptic feedback from their phone and handheld game controllers. Participants were compensated with 10$ for their time at the end of the study.

6.1.3 Procedure. The study was conducted in a controlled quiet environment. Participants sat down on a chair at a table with a desktop computer running the study software (see Figure 2, right). The study software was displayed on a large 24" Lenovo ThinkVision monitor. The haptic test apparatus was connected to a Macbook Pro (13", 2017) via Bluetooth. Participants always placed their left hand on the haptic dome to experience the haptic effects and operated the study software using a mouse in their right hand. They wore noise-canceling headphones (Sony WH-1000XM4) playing white noise at 15% volume to shield any audible sounds exerted from the actuator while still allowing them to listen to the scenario scene.

The task and rating criteria were explained to the participants in detail at the beginning of the study. Participants were able to familiarize themselves with the task in two practice trials comprising a baseline and foley_compression haptic effect sample for an immersive movie car crash scene. For each trial a scenario description was provided, like "Imagine you are in a virtual reality game and stroking a cat with your hand.". The description aimed to help participants put themselves into the specific application scenarios. The description was accompanied with a 3-5 seconds long video played in fullscreen for the corresponding scenario when pressing the play button. Participants could play the video up to 5 times. The haptic effect was played during the video based on the manually configured onset time and participants felt the haptic effect through their left hand while simultaneously watching the scenario video at the monitor.

For each trial participants first rated five questions regarding the experiential dimensions of the HAPTIC_EXPERIENCE on a 7-point Likert-scale (1 strongly disagree - 7 strongly agree). While Sathiyamurthy et al. [103] explored a first scale with multiple questions per dimension and find that the experiential dimensions can be measured through a survey, they do not recommend using their questions until a validated scale is produced. Since their 20 question instrument to access the different dimensions for each trial was too extensive for our study design (60 trials per participant), we decided to come up with a more concise set of questions (i.e. one question per dimension) which we confirmed in pilot tests to be meaningful to participants. These included: *Harmony:* "I think that the haptic effect fits well with the given context and other sensory outputs." *Autotelics:* "I think that the haptic effect feels good independent of the given context and other sensory outputs." *Expressivity:* "I think that the haptic effect forms a unique experience that allows me to easily differentiate it from other haptic effects." *Realism:* "The haptic effect convincingly portrays what I would expect to feel in reality." *Immersion:* "I felt fully immersed in the activity or experience." The average of these ratings provided a measure for the HAPTIC_EXPERIENCE from 1 (worst) to 7 (best). Participants furthermore rated DESCRIPTION_MATCHING "The haptic effect effectively conveys '[haptic effect description]'." on a 7-point Likert-scale (1 strongly disagree - 7 strongly agree).

6.1.4 Analysis. We performed repeated measures ANOVA and Holm correction for post-hoc tests. We analyzed comparisons for interactions between *scenario* × *haptic_effect_type* conditional on *scenario* to identify the impact of our pipeline components on different scenarios.

6.1.5 Results. The experiment took 41.10 minutes on average ($SD = 12.00$). We first report results with focus on the effect of HAPTIC_EFFECT_TYPES on different ratings, followed by the effect of SCENARIOS on different ratings. We display the overall rating scores in Figure 7.

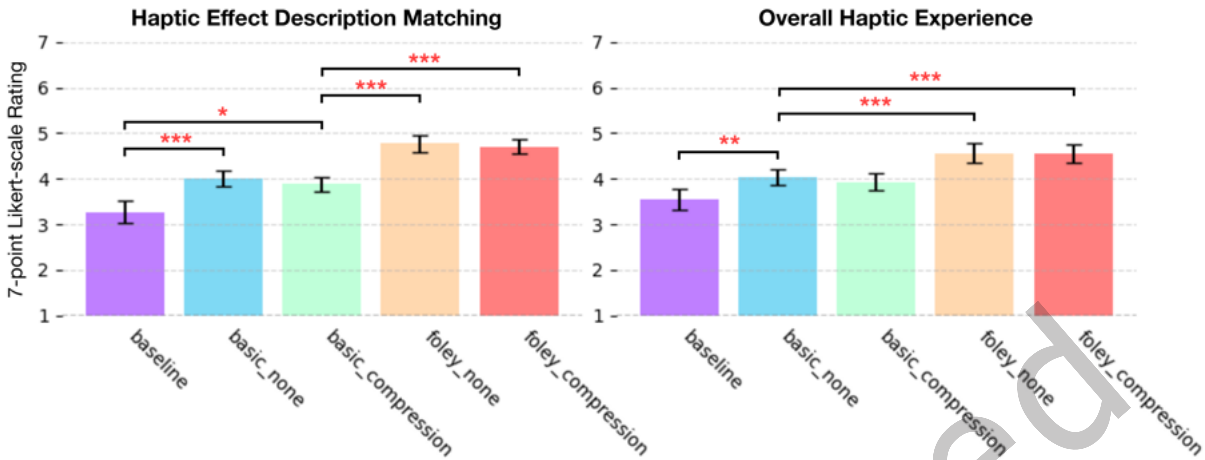

Fig. 7. **Pipeline Ablations Performance:** Mean DESCRIPTION_MATCHING and HAPTIC_EXPERIENCE ratings on a 7-point Likert-scale with 95% confidence intervals (higher ratings indicate better performance). Overall the mean scores were distributed between rating values of three and five. All generated haptic effects were rated better than baseline. Generated haptic effects with active *Foley Interpreter* were rated highest. The compression of our *Dynamic Audio Processor* did not show to effect mean ratings significantly.

Description Matching

DESCRIPTION_MATCHING was rated $M = 4.12$, $SD = 1.86$ on average. An ANOVA indicated a statistically significant main effect of HAPTIC_EFFECT_TYPE on DESCRIPTION_MATCHING ($F_{4,76} = 30.24$, $p < 0.01$), $\eta_p^2 = 0.61$. Post-hoc tests with Holm correction revealed a significant effect ($p < 0.05$) between all pairs, but {basic_compression, basic_none} ($p = 0.78$), and {foley_compression, foley_none} ($p = 0.78$). foley_none was rated highest ($M = 4.77$, $SD = 1.72$), followed by foley_compression ($M = 4.71$, $SD = 1.71$), basic_none ($M = 3.99$, $SD = 1.78$), basic_compression ($M = 3.88$, $SD = 1.79$), and baseline ($M = 3.27$, $SD = 1.88$). Cohen's d effect sizes for each pair depicted: {baseline, basic_compression} ($d = 0.34$), {baseline, basic_none} ($d = 0.40$), {baseline, foley_compression} ($d = 0.80$), {baseline, foley_none} ($d = 0.83$), {basic_compression, basic_none} ($d = 0.06$), {basic_compression, foley_compression} ($d = 0.46$), {basic_compression, foley_none} ($d = 0.49$), {basic_none, foley_compression} ($d = 0.42$), {basic_none, foley_none} ($d = 0.45$), {foley_compression, foley_none} ($d = 0.03$).

Overall Haptic Experience

Overall HAPTIC_EXPERIENCE was rated $M = 4.12$, $SD = 1.61$. An ANOVA indicated a statistically significant main effect of HAPTIC_EFFECT_TYPE on HAPTIC_EXPERIENCE ($F_{4,76} = 23.69$, $p < 0.01$), $\eta_p^2 = 0.56$,. Post-hoc tests with Holm correction revealed a significant effect ($p < 0.05$) between all pairs, but {baseline, basic_compression} ($p = 0.07$), {basic_compression, basic_none} ($p = 0.60$), and {foley_compression, foley_none} ($p = 0.90$). foley_none was rated highest ($M = 4.56$, $SD = 1.55$), followed by foley_compression ($M = 4.54$, $SD = 1.53$), basic_none ($M = 4.03$, $SD = 1.53$), basic_compression ($M = 3.93$, $SD = 1.56$), and baseline ($M = 3.54$, $SD = 1.63$). Cohen's d effect sizes for each pair depicted: {baseline, basic_compression} ($d = 0.24$), {baseline, basic_none} ($d = 0.30$), {baseline, foley_compression} ($d = 0.63$), {baseline, foley_none} ($d = 0.65$), {basic_compression, basic_none} ($d = 0.06$), {basic_compression, foley_compression} ($d = 0.39$), {basic_compression, foley_none} ($d = 0.40$), {basic_none, foley_compression} ($d = 0.33$), {basic_none, foley_none} ($d = 0.34$), {foley_compression,

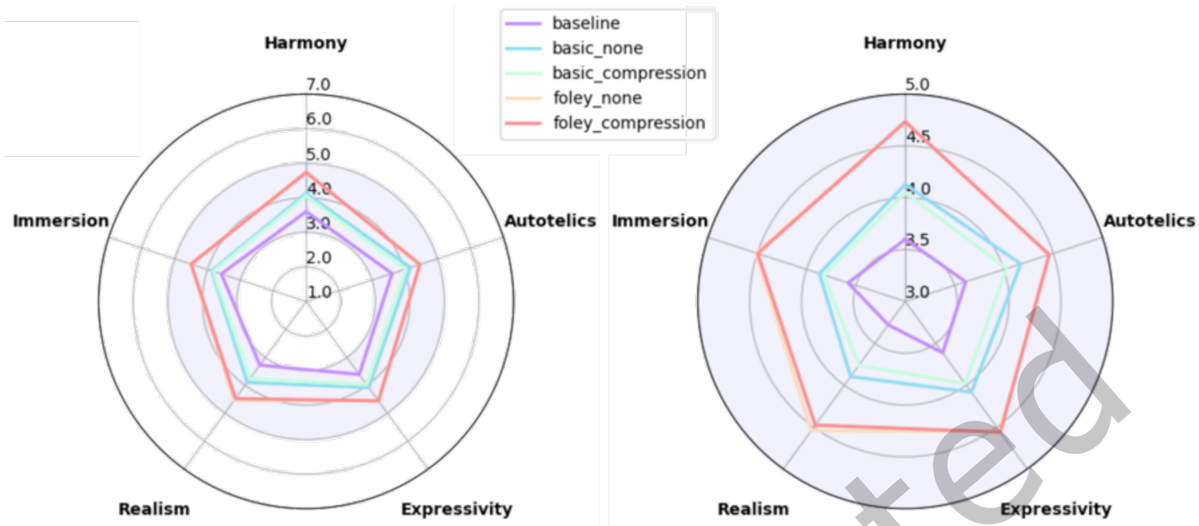

Fig. 8. **Experiential Dimensions Ratings:** The individual mean ratings of the five Experiential Dimensions of haptic experience for different pipeline components on a 7-point Likert-scale. **Left:** Full range plot from 1 to 7, revealing a narrow rating between 3 and 5. Overall foley conditions were rated higher than the basic conditions and baseline. **Right:** Partial plot highlighting differences within the range from 3 to 5. Slightly reduced realism ratings for the baseline condition becomes apparent.

foley_none} ($d = 0.01$).

Individual Experiential Dimensions

We display the individual ratings of the experiential dimension ratings in Figure 8.

Harmony: Overall HARMONY was rated $M = 4.25$, $SD = 1.71$. An ANOVA indicated a statistically significant main effect of HAPTIC_EFFECT_TYPE on HARMONY ($F_{4,76} = 29.14$, $p < 0.01$), $\eta_p^2 = 0.60$. Post-hoc tests with Holm correction revealed a significant effect ($p < 0.05$) between all pairs, but {basic_compression, basic_none} ($p = 0.80$), and {foley_compression, foley_none} ($p = 1.00$). foley_none was rated highest ($M = 4.73$, $SD = 1.55$), followed by foley_compression ($M = 4.73$, $SD = 1.62$), basic_none ($M = 4.13$, $SD = 1.68$), basic_compression ($M = 4.05$, $SD = 1.67$), and baseline ($M = 3.60$, $SD = 1.77$).

Autotelics: Overall AUTOTELICS was rated $M = 4.15$, $SD = 1.66$. An ANOVA indicated a statistically significant main effect of HAPTIC_EFFECT_TYPE on AUTOTELICS ($F_{4,76} = 10.33$, $p < 0.01$), $\eta^2 = 0.04$. Post-hoc tests with Holm correction revealed a significant effect ($p < 0.05$) between the pairs {baseline, foley_compression}, {baseline, foley_none}, {basic_compression, foley_compression}, and {basic_compression, foley_none}. For all other pairs no significant effect was found ($p >= 0.05$). foley_none was rated highest ($M = 4.46$, $SD = 1.64$), followed by foley_compression ($M = 4.46$, $SD = 1.65$), basic_none ($M = 4.17$, $SD = 1.62$), basic_compression ($M = 4.03$, $SD = 1.57$), and baseline ($M = 3.61$, $SD = 1.69$).

Expressivity: Overall EXPRESSIVITY was rated $M = 4.16$, $SD = 1.73$. An ANOVA indicated a statistically significant main effect of HAPTIC_EFFECT_TYPE on EXPRESSIVITY ($F_{4,76} = 13.04$, $p < 0.01$), $\eta^2 = 0.05$. Post-hoc

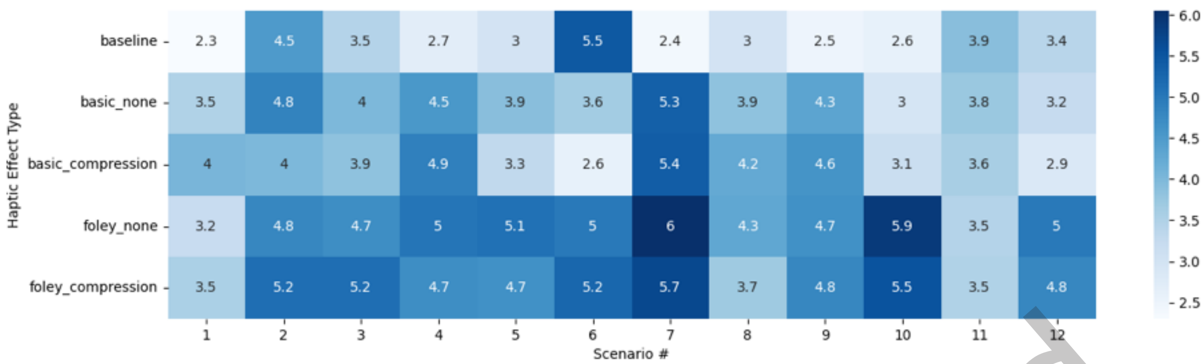

Fig. 9. **Individual Scenario Ratings:** Mean DESCRIPTION_MATCHING ratings (1 not matching at all, 7 perfectly matching) for each HAPTIC_EFFECT_TYPE × SCENARIO pair. Higher values indicate better ratings (displayed darker). See Section 6.1.1 and Section 1 for effect types and scenarios.

tests with Holm correction revealed a significant effect ($p < 0.05$) between all pairs, but {baseline, basic_compression} ($p = 0.24$), {basic_compression, basic_none} ($p = 0.79$), and {foley_compression, foley_none} ($p = 0.96$). foley_compression was rated highest ($M = 4.56$, $SD = 1.63$), followed by foley_none ($M = 4.55$, $SD = 1.69$), basic_none ($M = 4.08$, $SD = 1.66$), basic_compression ($M = 3.98$, $SD = 1.79$), and baseline ($M = 3.61$, $SD = 1.70$).

Realism: Overall REALISM was rated $M = 3.99$, $SD = 1.79$. An ANOVA indicated a statistically significant main effect of HAPTIC_EFFECT_TYPE on REALISM ($F_{4,76} = 30.82$, $p < 0.01$), $\eta^2 = 0.09$. Post-hoc tests with Holm correction revealed a significant effect ($p < 0.05$) between all pairs, but {basic_compression, basic_none} ($p = 0.46$), and {foley_compression, foley_none} ($p = 0.58$). foley_none was rated highest ($M = 4.54$, $SD = 1.74$), followed by foley_compression ($M = 4.48$, $SD = 1.71$), basic_none ($M = 3.90$, $SD = 1.71$), basic_compression ($M = 3.76$, $SD = 1.69$), and baseline ($M = 3.28$, $SD = 1.78$).

Immersion: Overall IMMERSION was rated $M = 4.06$, $SD = 1.80$. An ANOVA indicated a statistically significant main effect of HAPTIC_EFFECT_TYPE on IMMERSION ($F_{4,76} = 17.58$, $p < 0.01$), $\eta^2 = 0.05$. Post-hoc tests with Holm correction revealed a significant effect ($p < 0.05$) between all pairs, but {baseline, basic_compression} ($p = 0.44$), {baseline, basic_none} ($p = 0.12$), {basic_compression, basic_none} ($p = 1.00$), and {foley_compression, foley_none} ($p = 1.00$). foley_none was rated highest ($M = 4.51$, $SD = 1.75$), followed by foley_compression ($M = 4.50$, $SD = 1.72$), basic_none ($M = 3.87$, $SD = 1.73$), basic_compression ($M = 3.83$, $SD = 1.76$), and baseline ($M = 3.58$, $SD = 1.86$).

Individual Scenario Performance

The top rated scenario based on DESCRIPTION_MATCHING was Scenario #07 with an average rating $M = 4.98$, $SD = 1.90$ considering all pipeline components. The worst rating received Scenario #01 with $M = 3.32$, $SD = 1.68$. We display all DESCRIPTION_MATCHING ratings for the individual pipeline components in Figure 9.

An ANOVA indicated a statistically significant main effect of SCENARIO on DESCRIPTION_MATCHING ($F_{11,209} = 6.62$, $p < 0.01$), $\eta^2 = 0.07$. Post-hoc tests with Holm correction revealed a significant effect ($p < 0.05$) between all combinations of **Scenario #1** with {#10, #2, #4, #6, #7, #9}, all combinations of **Scenario #7** with {#10, #11, #12, #3, #5, #8}, and the pair **{#12, #2}**. For all other pairs no significant effect was found ($p >= 0.05$). Scenario #7 was rated highest ($M = 4.98$, $SD = 1.90$), followed by Scenario #2 ($M = 4.66$, $SD = 1.70$), Scenario #6 ($M = 4.39$,

$SD = 1.87$), Scenario #4 ($M = 4.38$, $SD = 1.95$), Scenario #3 ($M = 4.23$, $SD = 1.59$), Scenario #9 ($M = 4.18$, $SD = 1.60$), Scenario #10 ($M = 4.03$, $SD = 2.11$), Scenario #5 ($M = 3.99$, $SD = 2.00$), Scenario #12 ($M = 3.86$, $SD = 1.78$), Scenario #8 ($M = 3.82$, $SD = 1.79$), Scenario #11 ($M = 3.64$, $SD = 1.78$), and Scenario #1 ($M = 3.32$, $SD = 1.68$).

An ANOVA indicated a statistically significant main effect of SCENARIO on HAPTIC_EXPERIENCE ($F_{11,209} = 6.87$, $p < 0.01$), $\eta^2 = 0.06$. Post-hoc tests with Holm correction revealed a significant effect ($p < 0.05$) between all combinations of **Scenario #1** with {#2, #4, #7, #9}, all combinations of **Scenario #7** with {#10, #11, #12, #5, #8}, and the pairs **{#11, #2}**, {#11,#4}. For all other pairs no significant effect was found ($p >= 0.05$). Scenario #7 was rated highest ($M = 4.83$, $SD = 1.60$), followed by Scenario #2 ($M = 4.59$, $SD = 1.47$), Scenario #6 ($M = 4.41$, $SD = 1.58$), Scenario #4 ($M = 4.31$, $SD = 1.70$), Scenario #3 ($M = 4.27$, $SD = 1.36$), Scenario #9 ($M = 4.15$, $SD = 1.42$), Scenario #10 ($M = 4.06$, $SD = 1.79$), Scenario #12 ($M = 3.97$, $SD = 1.52$), Scenario #8 ($M = 3.84$, $SD = 1.61$), Scenario #5 ($M = 3.84$, $SD = 1.69$), Scenario #11 ($M = 3.64$, $SD = 1.59$), and Scenario #1 ($M = 3.53$, $SD = 1.45$).

An ANOVA indicated a statistically significant main effect of SCENARIO * HAPTIC_EFFECT_TYPE on DESCRIPTION_MATCHING ($F_{44,836} = 6.15$, $p < 0.01$), $\eta^2 = 0.14$. Post-hoc tests with Holm correction revealed a significant effect ($p < 0.05$) between the pairs {baseline, foley_compression}, {baseline, foley_none}, {basic_compression, foley_compression}, {basic_compression, foley_none}, {basic_none, foley_compression}, {basic_none, foley_none} for **Scenario #10**, the pairs {basic_compression, foley_none}, {basic_none, foley_none} for **Scenario #12**, the pair {baseline, foley_none} for **Scenario #4**, the pairs {baseline, basic_compression}, {basic_compression, foley_compression}, {basic_compression, foley_none}, {basic_none, foley_compression} for **Scenario #6**, the pairs {baseline, basic_compression}, {baseline, basic_none}, {baseline, foley_compression}, {baseline, foley_none} for **Scenario #7**), and the pairs {baseline, basic_compression}, {baseline, basic_none}, {baseline, foley_compression}, {baseline, foley_none} for **Scenario #9**. For all other pairs no significant effect was found ($p >= 0.05$).

An ANOVA indicated a statistically significant main effect of SCENARIO * HAPTIC_EFFECT_TYPE on HAPTIC_EXPERIENCE ($F_{44,836} = 5.90$, $p < 0.01$), $\eta^2 = 0.11$. Post-hoc tests with Holm correction revealed a significant effect ($p < 0.05$) between the pairs {baseline, foley_compression}, {baseline, foley_none}, {basic_compression, foley_compression}, {basic_compression, foley_none}, {basic_none, foley_compression}, {basic_none, foley_none} for **Scenario #10**, the pair {basic_none, foley_none} for **Scenario #12**, the pair {baseline, foley_none} for **Scenario #4**), the pairs {baseline, basic_compression}, {basic_compression, foley_compression} for **Scenario #6**, the pairs {baseline, basic_compression}, {baseline, basic_none}, {baseline, foley_compression}, {baseline, foley_none} for **Scenario #7**, and the pairs {baseline, basic_compression}, {baseline, basic_none}, {baseline, foley_compression}, {baseline, foley_none} for **Scenario #9**. For all other pairs no significant effect was found ($p >= 0.05$).

6.2 Study 2: Expressiveness & Pipeline Robustness

The objective of our second investigation with participants was to test if (1) our pipeline produces unique haptic effects that are mostly effective for the target scenario (expressiveness), and (2) our pipeline can generate consistent haptic effects that can enhance a specific application scenario (robustness). This study was approved by the institutional review board at the National University of Singapore (NUS-IRB-2024-570).

6.2.1 Design. We conducted a within-subjects study design. The study had two sessions with one task each, a ROBUSTNESS-TASK and a EXPRESSIVENESS-TASK.

The design for the ROBUSTNESS-TASK was 2 SCENARIOS × (6 FIXED DESCRIPTIONS + 6 VARIED DESCRIPTIONS) × 10 PARTICIPANTS = 240 samples in total. 24 trials per participant. We chose the first two scenarios (Scenario #1 and Scenario #2) from our focus group to test the robustness of our pipeline as we had at least six varied descriptions for each available. For each scenario we once generated six haptic effects using our full pipeline (foley_compression) based on the single most representative haptic effect description (see Section 4) (fixed descriptions). Additionally we generated six haptic effects for each scenario based on the top six haptic effect descriptions from our haptic effect elicitation survey (varied descriptions). Measured was the HAPTIC_EXPERIENCE, as well as the DESCRIPTION_MATCHING on a 7-point scale (1 worst, 7 best) as in study 1.

In the EXPRESSIVENESS-TASK we focused on all 6 SCENARIOS from the focus group. We used the same six full pipeline haptic effects (foley_compression) generated for the individual scenarios from study 1. For each scenario, participants performed A, B comparisons between all 6 HAPTIC EFFECTS of the different scenarios and were asked to choose the best matching for the displayed scenario.

Trials were fully randomized. Half of the participants first performed the ROBUSTNESS-TASK, half first performed the EXPRESSIVENESS-TASK. The tasks were conducted in two separate sessions and hold on two separate days. Participants took a break every 15 trials. Actuator noise was shielded. The study apparatus was kept consistent across all application scenarios as in study 1.

6.2.2 Participants. We recruited 10 participants aged $M = 30.70$ years ($SD = 6.73$), 5 female. One was left-handed. Seven reported basic experience with haptic feedback from their phone and handheld game controllers. Participants were compensated with 10$ for their time at the end of the study.

6.2.3 Procedure. The study setup was as in study 1. After filling the consent form and demographics, participants were assigned either the ROBUSTNESS-TASK or EXPRESSIVENESS-TASK first.

Participants were able to familiarize themselves with the task in test trials comprising a baseline and foley_compression haptic effect for an immersive movie car crash scene. Participants could read the description and watch the scenes as in study 1. For the ROBUSTNESS-TASK the participants performed the rating exactly as in study 1. For the EXPRESSIVENESS-TASK, participants could play the same scene using two different buttons. One button would play the scene with haptic effect A, the other one with haptic effect B. Participants were asked to select the effect that they felt matches better with the displayed scenario.

6.2.4 Analysis. For robustness we first calculated the mean standard deviation of the individual ratings for each scenario and participant. Taking the average of all mean standard deviations across participants resulted in a measure of the overall dispersion. A small dispersion of the ratings was interpreted as high robustness. Repeated measures ANOVA with Holm correction for post-hoc tests was carried out. For expressiveness we calculated the average times a haptic effect was ranked first for a given scenario (MATCH_SCORE). Played effects were grouped into EFFECT_GROUP with effects {designed_for_scenario, not_designed_for_scenario}. The more often haptic effects are ranked first for the scenario they were intended for, the more expressive the system is.

6.2.5 Results. Completing the Robustness and EXPRESSIVENESS-TASKS took $M = 15.36$ ($SD = 5.89$) and $M = 25.74$ ($SD = 7.48$) minutes respectively.

Robustness

An ANOVA revealed no statistically significant effect of SCENARIOS, DESCRIPTION TYPES (fixed or varied), or SCENARIOS × DESCRIPTION TYPES on the average dispersion of various measures (all $p >= 0.05$).

DESCRIPTION_MATCHING had the highest dispersion for varied descriptions with an average standard deviation of $M = 1.36$ ($SD = 0.64$), followed by fixed descriptions with $M = 1.18$ ($SD = 0.56$). HAPTIC_EXPERIENCE had the highest dispersion for varied descriptions with an average standard deviation of $M = 1.09$ ($SD = 0.53$), followed by fixed descriptions with $M = 0.87$ ($SD = 0.49$). HARMONY had the highest dispersion for varied descriptions with an average standard deviation of $M = 1.21$ ($SD = 0.43$), followed by fixed descriptions with $M = 0.95$ ($SD = 0.48$). AUTOTELICS had the highest dispersion for varied descriptions with an average standard deviation of $M = 1.19$ ($SD = 0.57$), followed by fixed descriptions with $M = 0.99$ ($SD = 0.50$). EXPRESSIVITY had the highest dispersion for varied descriptions with an average standard deviation of $M = 1.17$ ($SD = 0.51$), followed by fixed descriptions with $M = 1.10$ ($SD = 0.49$). REALISM had the highest dispersion for varied descriptions with an

Fig. 10. **Robustness of Repeated Generations:** The average standard deviation of individual participant ratings (including 95% confidence intervals) with respect to repeated generations based on fixed descriptions and varied descriptions as a measure of robustness (left). Confusion matrix of matching haptic effects to individual scenarios (right). Scores show how many times (as fraction) participants ranked a *Scenario* × *Haptic Effect* pair as best matching.

average standard deviation of $M = 1.33$ ($SD = 0.62$), followed by fixed descriptions with $M = 1.05$ ($SD = 0.49$). IMMERSION had the highest dispersion for varied descriptions with an average standard deviation of $M = 1.23$ ($SD = 0.60$), followed by fixed descriptions with $M = 1.04$ ($SD = 0.48$). The average standard deviation values for DESCRIPTION_MATCHING and HAPTIC_EXPERIENCE between fixed and varied descriptions can be found in Figure 10 (left).

Expressiveness
An ANOVA indicated a statistically significant main effect of EFFECT_GROUP on MATCH_SCORE ($F_{1,9} = 14.05$, $p < 0.01$), $\eta_p^2 = 0.61$. A post-hoc test revealed a statistically significant higher MATCH_SCORE of effects designed_for_scenario ($M = 0.43, SD = 0.23$) compared to effects not_designed_for_scenario ($M = 0.11, SD = 0.05$) ($p < 0.01$), Cohen's $d = 1.92$. No other significant main effect of SCENARIO or SCENARIO × EFFECT_GROUP on MATCH_SCORE was found ($p >= 0.05$). For Scenario #01 the top matched haptic effect was effect #01 (5 times). For Scenario #02 the top matched haptic effect was effect #06 (6 times). For Scenario #03 the top matched haptic effect was effect #03 (5 times). For Scenario #04 the top matched haptic effect was effect #04 (4 times). For Scenario #05 the top matched haptic effect was effect #05 (4 times). For Scenario #07 the top matched haptic effect was effect #07 (7 times). A confusion matrix with match scores for the individual scenario × haptic effect pairs can be found in Figure 10 (right).

7 Technical Analysis

To yield further insights on the effectiveness of our pipeline components from a technical standpoint, we performed an additional technical analysis with the, to our knowledge, most suitable available evaluation metrics. As some of the metrics adapted from the audio domain can be sensitive to structure and semantics of audio clips, trained on audio data, part of the results should be interpreted with caution. We highlight these cases in the following sections. We are not aware of any standardized technical evaluation metric specific to the haptic domain which would have been more applicable to assess the effectiveness of our pipeline.

7.1 Technical Analysis 1: Translating Haptic Descriptions to Audio Descriptions

The goal of our *Foley Interpreter* is to translate haptic effect descriptions to audio effect descriptions and we explicitly instruct the employed LLM to produce audio captions. TTA models such as the one used by our *Text-to-Audio Generator* are trained on large audio caption datasets similar to the widely used AudioCaps dataset [51].

To gain a technical measure on the effectiveness of our *Foley Interpreter* component, we therefore, first calculated the similarity between the haptic effect descriptions from our application scenario test set (section 4) and 2000 audio captions from the AudioCaps dataset. We then also calculated the similarity between the audio effect descriptions produced by our *Foley Interpreter* and the captions from AudioCaps.

A higher similarity of the output of our *Foley Interpreter* to the AudioCaps audio captions compared to the haptic effect descriptions used as input, would therefore indicate that our *Foley Interpreter* can fulfill part of its functionality. A measure of similarity was obtained by first generating vector embeddings for all haptic effect descriptions and audio captions using the OpenAI embeddings API, followed by computing the cosine similarity between the resulting vectors.

A two-sided t-test revealed a statistically significant difference between the means of the similarity scores of the audio effect descriptions ($M = 0.29, SD = 0.09$) and the haptic effect descriptions ($M = 0.25, SD = 0.07$) with $t(12) = 44.84$, $p < 0.01$. Our results, hence, indicate that our *Foley interpreter* indeed translates haptic effect descriptions to audio effect descriptions that are closer to the training sets of TTA models, like AudioCaps.

To provide an additional angle on the ability of our *Foley Interpreter* to preserve the semantic meaning of the described haptic experience within the produced audio descriptions from experts, we asked eight individuals aged $M = 38.13$ ($SD = 4.91$, all male) and experienced in designing haptic feedback, to compare and rate audio effect descriptions. To test whether the semantic adaption changes for other domains we furthermore included six scenarios from the medical, accessibility, and sports domain (see appendix C). All scenarios were rated on a 7-point Likert scale regarding the Overall Semantic Alignment between audio and haptic effect description, i.e. *"How well does the Audio Effect Description capture the meaning and qualities of the Haptic Effect Description?"* (1 not at all - 7 perfectly). All scenarios were rated on average $M = 4.59$ ($SD = 0.63$). Our initial 12 scenarios were rated $M = 4.78$ ($SD = 0.57$), and the additional 6 cross validation scenarios $M = 4.21$ ($SD = 0.61$). A t-test did not show a statistically significant difference between both sets, with a low p-value ($t(37) = 1.91$, $p = 0.06$). These results indicate that experts judge our *Foley Interpreter* to be able to preserve the semantic meaning in the produced audio description moderately well, whereby cross validation scenarios from the medical, sports and accessibility domains appeared to perform slightly worse.

7.2 Technical Analysis 2: Robustness

The aim of our second analysis was to gain a first objective measure on how robustly our pipeline could generate haptic effects in a series when (1) keeping the input description fixed for a single scenario ("Fixed Description"), and (2) when using varied descriptions for the same scenario ("Varied Description"). Here we focused on the six scenarios from our focus group (section 4), as we had at least six different haptic effect descriptions available for each scenario.

We calculated the audio mean similarity scores between the generated audio outputs based on PANN [55] embeddings for different pipeline ablations (deactivation of different components), to identify the effect of individual components on the robustness of generating haptic effects. The ablations were the same as for the human evaluation in Section 6.1.1.

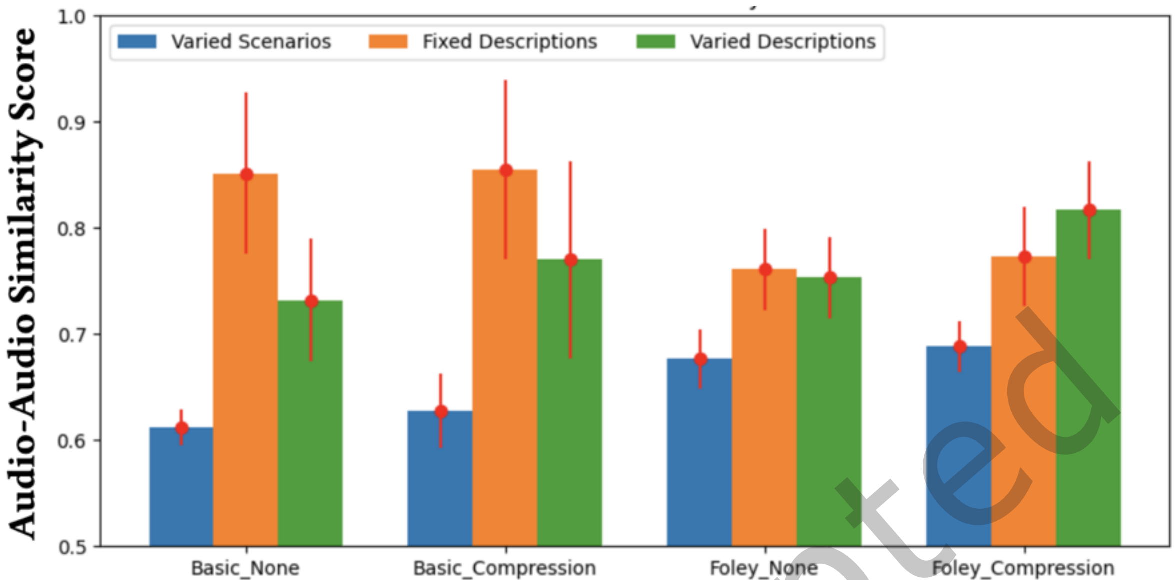

Fig. 11. **Technical Analysis 2 (Robustness):** Bar plots showing mean audio-audio similarity scores (with standard deviation) between audio samples generated based on *Varied Scenarios*, *Fixed Descriptions* and *Varied Descriptions* for different ablation types of our pipeline. Similarity scores from 0 - 1.0. Higher scoring indicate higher similarity. As expected, *Varied Scenarios* performed worst, as we compared descriptions from different scenarios. *Varied Descriptions* for the same scenario as well as repeated generations for a *fixed description* scored significantly better.

As a baseline to compare to, we chose to calculate the same measures when using the haptic effect descriptions of different scenarios (”Varied Scenarios”). Naturally we would expect lower similarity scores here as the descriptions would focus on different, not the same scenario. The results are displayed in Figure 11. For each pipeline ablation with ”Varied Scenarios”, ”Fixed Descriptions”, ”Varied Descriptions” we plot the similarity scores with standard deviations.

A one-way ANOVA between the four ablations of the pipeline for all three measures of *‘Varied Scenarios’* ($F_{3,20} = 9.57$, $p < 0.01$), *‘Fixed Descriptions’* ($F_{3,68} = 9.09$, $p < 0.01$), and by *‘Varied Descriptions’* ($F_{3,68} = 10.35$, $p < 0.01$) shows significant differences between mean audio similarity scores.

For *‘Varied Scenarios’*, foley_compression scored the highest ($M = 0.69$, $SD = 0.02$), followed by foley_none ($M = 0.67$, $SD = 0.03$), basic_compression ($M = 0.63$, $SD = 0.03$), and basic_none ($M = 0.61$, $SD = 0.02$). For *‘Fixed Descriptions’*, both basic_none ($M = 0.85$, $SD = 0.08$) and basic_compression ($M = 0.85$, $SD = 0.08$) scored highest, followed by foley_compression ($M = 0.77$, $SD = 0.07$), and foley_none ($M = 0.76$, $SD = 0.04$). For *‘Varied Descriptions’*, foley_compression ($M = 0.81$, $SD = 0.05$) scored highest, followed by basic_compression ($M = 0.77$, $SD = 0.09$), foley_none ($M = 0.75$, $SD = 0.06$), and basic_none ($M = 0.73$, $SD = 0.06$).

‘Varied Scenarios’ was significant lower than *‘Fixed Descriptions’* for all pipeline ablations, indicating that the generation with *‘Fixed Descriptions’* is significant more robust with a large difference and high similarity score around 85% for basic ablations without *Foley Interpreter* component. Similarly *‘Varied Descriptions’* scored higher than *‘Varied Scenarios’*, and worse than *‘Fixed Scenarios’* for basic ablations. The increase of similarity scores for *‘Varied Scenarios’* for ablations that include the *Foley Interpreter* indicate that the *Foley Interpreter* module translated haptic effect descriptions to a smaller semantic language space. At the same time the similarity scores

for *'Varied Descriptions'* and *'Fixed Descriptions'* for fixed descriptions decrease significantly, indicating lower robustness for repeated generations of with *'Fixed Descriptions'*. One explanation could be that the produced audio descriptions by the *Foley Interpreter* are longer and contain vocabulary more similar to the training set of the TTA model. It therefore can more easily trigger different rich sound effects, resulting in a lower robustness but potentially increasing expressiveness and the effectiveness of conveying haptic effects. While we hope the scores will serve as benchmark for future work, we highlight, that interpreting the similarity scores has to be done cautiously since the metric is generally used for sound effects not haptic effects. Sound effects comprise audio clips with a frequency spectrum from 0-8kHz while our generated haptic effect clips have a frequency spectrum from 0-1kHz.

7.3 Technical Analysis 3: Expressiveness

The goal of the third analysis was to get an objective measure on how well our pipeline can generate effects that only reflect the haptic effect description used for a specific scenario. The more the output reflects the haptic effect description of the intended scenario, but not the description of other scenarios, the higher the expressiveness of our system.

To measure how well the output reflects the input description we used CLAP [134] that allowed us to compute audio-text embeddings from the haptic effect descriptions as well as the generated audio samples. CLAP is a pre-trained model jointly trained on a large audio dataset and its associated text captions. For technically evaluating our method, we compute the cosine similarity between the text embeddings and audio embeddings generated by CLAP. A higher text-audio cosine similarity score indicates that the generated sounds are representative of the semantic attributes within the textual description.

Also here we calculated the scores for different ablations of our pipeline and focused on the six scenarios from the focus group. Figure 12 shows the confusion matrices with similarity scores of all pairwise combinations between audio generations and haptic effect descriptions for the scenarios. We calculated the mean similarity scores for the matching group types of scenarios where the haptic effect matched the scenario ("matching group") and the group where haptic effects did not match their scenario ("not matching group").

A two-way ANOVA of matching group type and ablation type on the similarity score revealed a significant main effect of matching group type, $F_{1,136} = 92.58$, $p < 0.01, \eta^2 = 0.40$. The matching group revealed significantly higher similarity scores ($M = 0.22$, $SD = 0.07$) compared to the not matching group ($M = 0.12$, $SD = 0.04$). We also found a significant main effect of ablation type on the similarity score $F_{3,136} = 3.37$, $p = 0.02$, $\eta^2 = 0.07$. Pairwise comparisons showed significantly lower scores in the basic_compression group compared to foley_none group, $p < 0.05$. The mean similarity score for matching haptic effects for basic_compression ($M = 0.18$, $SD = 0.08$) was significantly higher compared to basic_compression ($M = 0.10$, $SD = 0.04$) , $p < 0.05$.

Overall, the group of haptic effects matching the target scenario were rated higher, indicating our pipeline can produce expressive samples for different applications using haptic effect descriptions. While we could only find a significant effect between basic_compression and foley_none for the ablation type comparisons, the confusion matrices let suggest a general trend of increased similarity scores around the diagonal for ablations with *Foley Interpreter*. This indicates a potential higher expressiveness when using the *Foley Interpreter* component. However, the overall scoring of our full pipeline (foley_compression) were still low ($M = 0.24$, $SD = 0.05$) suggesting there is significant room to improve. We note again, that the applied similarity measure has not yet been verified on the applicability for haptic effect samples that consist of heavily processed audio samples.

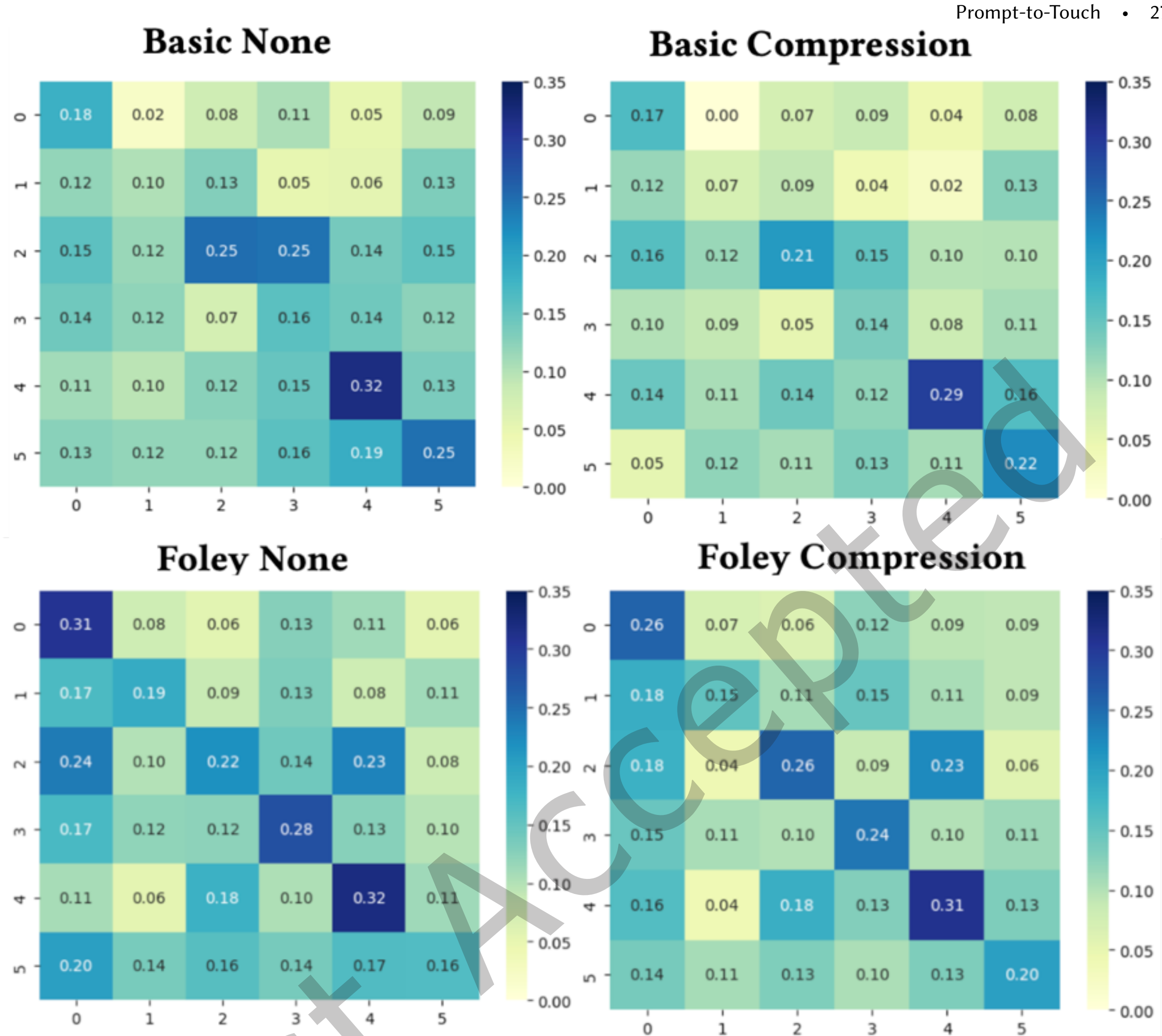

Fig. 12. **Technical Analysis 3 (Expressiveness):** Confusion matrices with text-audio similarity scores of all pairwise combinations between haptic effect descriptions and audio generations for six scenarios from the focus group for each ablation type. Similarity scores from 0 - 1.0. Higher scoring indicate higher similarity.

8 Discussion

8.1 Enhancing Haptic Experience in Different Applications Through Expressive Haptic Effects

Overall we could observe that our pipeline was able to enhance the haptic experience ($M = 4.54$, $SD = 1.53$) compared to the baseline ($M = 3.54$, $SD = 1.63$) in 10 out of 12 different scenarios from 5 different application areas. Higher effect sizes for description matching, as well as participants higher ranking of haptic effects in context of their intended scenario, show that this was not due to randomly generated dynamic noise, but due to our pipeline generating haptic effects specific to their description and better fitting to their intended target application. Only in two scenarios baseline performed equal or better than generated haptic effects. In the Industrial Application scenario, where the haptic effect was meant to resemble an alarm (Scenario #6), the baseline vibration was able to enhance the haptic experience similarly well as our generated effect ($M = 5.09$,

$SD = 1.06$ vs. $M = 5.09$, $SD = 1.23$). In the Social Communication scenario (Scenario #11) our generated effect entirely failed to provide a meaningful haptic experience that reflects a happy message ($M = 3.26$, $SD = 1.46$). We provide a potential explanation in Section 8.2.1 with respect to our *Text-to-Audio Generator* component.

8.1.1 Effective Dynamic and Organic Effects for Immersive Experiences. For some types of scenarios, such as Immersive Experiences, the generated effects were generally rated higher compared to others. The highest rated haptic effects regarding the haptic experience as well as the description matching were the rain effect (Scenario #7, $M = 5.38$, $SD = 1.25$) and bird pecking (Scenario #10, $M = 5.29$, $SD = 1.39$). Here our pipeline was able to generate effects that match the expected characteristics for the target haptic experience well in accordance with the audio-visual scene. 8 of 20 participants verbally indicated that they liked the rain effect most. A common theme for effects that worked well, appeared to be dynamic and organic patterns. In line with this, many participants (9 of 20) also highlighted that they liked the effect played for touching and moving their hand along a virtual rock: "It really feels like touching a rock. [..] It moves slower and faster.. that makes it pretty good." (P12) (Scenario #9, $M = 4.56$, $SD = 1.52$). He referred to the characteristic scratching texture rendered with rising and falling intensity.

8.1.2 Rendering Gentle Effects. Stroking the soft fur of a cat (Scenario #1) depicted one immersive scenario where our pipeline was not able to generate a haptic effect well. Here baseline ($M = 2.79$ $SD = 1.10$) as well as our generated effect ($M = 3.93$, $SD = 1.40$) scored low. 8 of 20 participants expressed that they did not feel the effect for the cat scenario worked well. Despite this, participants tended to still rank this haptic effect as best matching to the cat scenario in study 2. This aligns with some participants stating that they felt the haptic effect for the cat scenario carried some of the expected characteristics of stroking a cats fur, but the haptic effect was rendered too intensely to feel like a realistic stroking effect. However, when looking at the generated audio effect description "Soft, delicate swooshing with gentle whispering touches, mimicking the feel of a cat's fur brushing lightly against the skin.", several key words appear to highlight the gentleness of the haptic effect such as "delicate", "gentle", "lightly". This could indicate that it is yet difficult for our pipeline to adequately reflect the appropriate intensity of the haptic effect based on the effect description, specifically when it comes to very light and gentle effects.

8.1.3 Synchronizing Steady and Rhythmic Patterns. Application scenario #5 revealed one of the strongest discrepancies between haptic experience ($M = 4.29$, $SD = 1.80$) and description matching ($M = 4.65$, $SD = 1.87$). Here the haptic effect was supposed to render the rhythmic pattern of a metronome. While the haptic effect was rendering in a rhythmic pattern close to a metronome, it was rendered slightly out of sync with other sensory outputs. This was easily noticed by participants and decreased the haptic experience ratings particularly with respect to the harmony dimension: "The metronome is good if you get the right fit." (P11). In contrast, participants were much more forgivable for slightly misaligned knob clicks rendered in Scenario #3. Here, a virtual knob was rotated dynamically in VR. While the click events in both scenarios were of similar nature, the key difference was that one was steady and rhythmic, where as the other effect entailed more random elements. This exemplifies, the current limitation of our pipeline to generate haptic effects that synchronize appropriately with audio visual content.

8.2 The Effectiveness of Different Pipeline Components

8.2.1 Text-to-Audio Generator. Generating audio effects dynamically based on the haptic description using the *Text-to-Audio Generator* and playing them directly as haptic feedback (basic_none, $M = 3.99$, $SD = 1.78$) outperformed the static baseline ($M = 3.27$, $SD = 1.88$) ($p < 0.05$). However, the mean subjective ratings for haptic description matching of basic_none samples remained around a Likert-scale value of four, indicating that participants were unsure whether the generated effects match the description. In 9 of 12 scenarios participants rated auto generated effects higher than baseline. In only three scenarios the description matching was rated

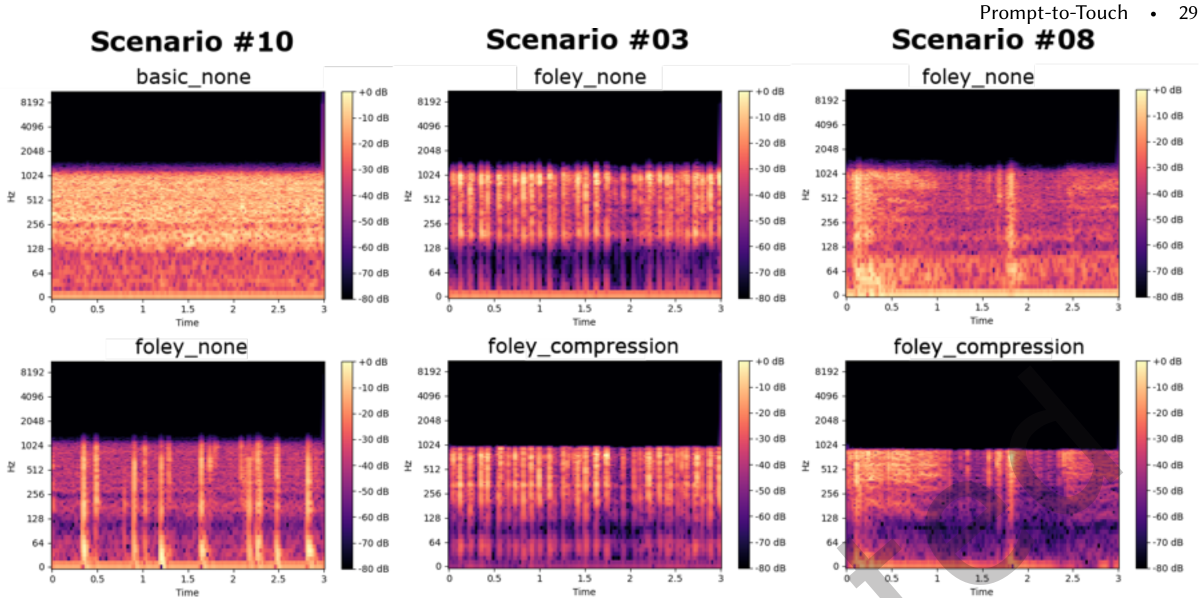

Fig. 13. **Component Performance Examples:** The *Foley Interpreter* successfully translated the haptic effect description into a sound effect description highlighting the key characteristics for Scenario #10 (left). Scenario #03 demonstrates a case where the *Dynamic Audio Processor* was able to shift the key characteristics into the lower frequency spectrum (center). In several cases like Scenario #08, the *Dynamic Audio Processor* failed to shift the key characteristics into the lower frequency spectrum and even resulted in less effective samples (right).

greater or equal a Likert-scale value of 4.50. Directly using the auto generated sample without involving other components of our pipeline was most effective for the rain scenario (Scenario #7, $M = 5.35$, $SD = 1.73$). We explain this due to the sound of rain (1) being a common sound effect that is (2) often characterized by hearing the events of raindrops falling on the roof of a building. Due to the latter, many audio recordings are already characterized by low frequency effects. In this case, the same events are anticipated for the haptic experience. As the events are already present in the lower frequency spectrum of the raw audio effect, it allows rendering them effectively in their unprocessed form. Similarly the increased description matching ratings for the sounds of water flowing into a glass (Scenario #2, $M = 4.85$, $SD = 1.81$), or a beating heart (Scenario #4, $M = 4.55$, $SD = 1.70$) could be explained. These, therefore, depict good examples for the intersection of the audio and haptic experience spaces as depicted in Figure 3. In contrast, abstract haptic effect descriptions like "[..] happy sensations on the skin [..]" (Scenario #11, $M = 3.75$, $SD = 1.80$) and "long and strong pulses" (Scenario #6, $M = 2.60$, $SD = 1.57$) did not work well. In fact, the latter scored significantly worse than baseline ($M = 5.45$, $SD = 1.47$). Here, the sound generation appeared failing to capture essential characteristics such as "pulsing", and instead generated noisy rumbling. Participants appeared to clearly prefer the clean, strong and long sensation exhibited by the baseline sample. These observations support the idea that our *Audio Effect Generator* and the used TTA model is more effective in generating realistic natural effects compared to artificial and abstract effects.

8.2.2 Foley interpreter. When including the *Foley Interpreter* component in the pipeline, participants started tending towards agreeing that the generated haptic effects match to their description well (foley_none, $M = 4.77$, $SD = 1.72$) compared to basic_none ablations without *Foley Interpreter* ($M = 3.99$, $SD = 1.78$) ($p < 0.05$). For many scenarios (10 out of 12) the *Foley Interpreter* further increased the description matching rating by translating the haptic description to an appropriate audio description. For example, in Scenario #10 the initial haptic description "feel a bird pecking lightly on the skin" focuses on an experience that does not involve an audible experience. The basic_none generation resulted in a noisy sample with a description matching rating of $M = 2.95$, $SD = 1.73$. In contrast, the foley interpreted description ”Soft, high-pitched taps in a gentle, irregular rhythm, resembling a bird lightly pecking on a surface.” clearly highlighted on audible characteristics that were also reflected in the generated sample (foley_none, $M = 5.90$, $SD = 1.07$), and visible in the spectrogram (see Figure 13). Similarly the foley interpreter turned ”taps on the skin” (basic_none, $M = 3.90$, $SD = 1.83$) in Scenario #5 into ”taps on a wooden table” (foley_none, $M = 5.10$, $SD = 1.68$). The effectiveness of translating haptic effect descriptions into audio effect descriptions was furthermore supported by the results of our technical analysis (section 7.1), which indicate that the produced audio descriptions were more similar to audio captions from training datasets used for TTA models as the one we employed in our *Text-to-Audio Generator*.

The *Foley Interpreter* component not only appeared to translate non audible haptic effect descriptions into audible ones, but could also sometimes successfully highlight key characteristics of the haptic experience. For example in Scenario #12 the haptic description matching rating likely increased from $M = 3.25$ ($SD = 1.62$) (basic_none) to $M = 5.00$ ($SD = 1.56$) (foley_none), because the generated audio based on the initial haptic description (without *Foley Interpreter*) did not reflect the characteristic ”fading away” well. In contrast, the audio description generated by the *Foley Interpreter* stressed multiple times on this key aspect by using the phrases “rushes out” and “quickly fading”.

Not in all cases the *Foley Interpreter* managed to fulfill its functionality. In Scenario #1, it failed to translate the haptic effect description ”The soft fur softly rubbing against the skin corresponding to the cat’s fur.” into a fully audible description. Here, the audio description still focused on the inaudible haptic events on the skin ”Soft, delicate swooshing with gentle whispering touches, mimicking the feel of a cat’s fur brushing lightly against the skin.”. In most of the cases the *Foley Interpreter* functioned as intended, but examples like Scenario #1 highlight opportunities to further improve the Foley Interpreter component by ensuring it translates haptic experiences into the audio space more reliably. As prior work suggests the use of longer descriptions to increase the effectiveness of TTAs[7] and repeated paraphrasing of key characteristics of the haptic effect in Scenario #12 indeed appeared to improve the resulting haptic effect, future modification could aim to tune the *Foley Interpreter* to actively exploit this technique.

8.2.3 Dynamic Audio Processor. Overall, the Dynamic Audio Processor did not show a significant effect on the haptic experience and description matching ratings. Opposing to our assumption that the dynamic compression step would only positively effect samples, especially when the key characteristic of the generated sound is located in the frequency spectrum beyond 1kHz, in several cases (12/24, considering comparisons between basic_none and basic_compression, as well as foley_none and foley_compression) the ratings would be lower after dynamic audio processing was applied. This was particularly the case for Scenario #5, #6, and #8. For Scenario #5 the Dynamic Audio Processor appeared to be able to shift some of the higher frequency characteristics into the lower level. However, it also appeared to introduce other side effects. The generated sound effect that aimed to resemble a ”Steady, crisp tap sound, like gentle finger taps on a wooden table at a rhythmic pace.” (foley_none) turned from relatively crisp taps into more mushy prolonged taps with slight rumbling which was also evident in the spectrogram around 0-50Hz. Prior work aiming to shift audio characteristics to the lower frequency spectrum for haptic feedback reported similar effects like ”muffling” emerging after processing. Also in Scenario #8 we could observe a negative side effect of our processing technique (Figure 13, right). Here the compression failed to translate some of the key characteristics to the haptic perceivable spectrum and instead introduced a void around 0-300Hz. This likely resulted in a lower rating for the foley_compression sample as dominant features felt too crisp and broken. In this case the original foley_none sample matched the haptic effect description of ”[..] a paper bouncing on your skin like it is thrown into a trashcan.” better compared to with compression as it felt softer and less artificial.

[7]https://huggingface.co/spaces/declare-lab/tango.

Nevertheless, in some scenarios as in Scenario #3 (Figure 13, center) the *Dynamic Audio Processor* appeared to be able to fulfill its intended functionality by carrying the characteristics into the low frequency spectrum successfully. We believe improving the *Dynamic Audio Processing* step could significantly elevate the haptic experience and allow producing more effective haptic effects that match the haptic description. Based on observations from empirical tests and user studies, we believe one way to improve the dynamic audio processing could be to focus on compressing sounds to a lower frequency spectrum around 0-500Hz instead of 0-1000Hz, so that the most characteristic part of the sound is translated to the lower haptic spectrum which may be more easily perceivable. This is the range many actuators are tuned towards and two of the key mechanoreceptors – responsible to sense tactile stimuli – are more sensitive to. Here, the Pacinian Corpuscles are known to respond particularly to vibrations around 250Hz, and Meissner corpuscles are known to particulary respond to dynamic deformations at a frequency around 40Hz [17, 69, 92]. With respect to this, the post processing steps could try to take into account these perceptual characteristics further, by e.g. adjusting haptic effects to the response characteristics of the Pacinian Corpuscles and Meissner Corpuscles. This could help rendering audio samples more accurately to the user as haptic effects.

8.2.4 Foley Interpreter vs. Dynamic Audio Processor. We believe different dynamics between the *Foley Interpreter* and *Dynamic Audio Processing* components could be explored in the future to produce haptic effects more effectively. On one hand the Foley Interpreter could be optimized to already translate haptic effect descriptions to ones that only produce sounds in the lower frequency spectrum, hence not requiring additional processing that shifts them to the lower spectrum and preventing introducing unwanted "muffling" or other side effects. On the other hand, if the *Dynamic Audio Processor* would work effectively, the Foley Interpreter could potentially be tuned to focus more on producing rich sound effects in the upper spectrum that carry important characteristics for the haptic effect. If the *Foley Interpreter* would understand precisely what types of sounds translate well to haptic effects, this technique could be exploited more effectively.

8.3 Robustness of Generating Haptic Effects

Generally our pipeline appeared to be robust in enhancing the haptic experience across different applications (10 out of 12). When looking at repeated generations for the same scenario, our technical analysis and preliminary user study indicated a higher robustness for *fixed descriptions* and *varied descriptions*. Participants rated description_matching and overall_rating relatively consistent with a lower average standard deviation around 1.00. Only the description_matching rating for varied descriptions exposed an elevated deviation, indicating less robustness. Compared to our technical analysis the results of our user study indicate a trend of varied description ratings being less robust compared to fixed description ratings which is more consistent with our expectations. In contrast, in our technical analysis varied descriptions scored slightly higher in similarity compared to fixed descriptions. In addition, the *Foley Interpreter* appeared to reduce robustness in our technical analyis. However, at the same time, based on other measures, the *Foley Interpreter* showed to improve the expressiveness of our pipeline and allowed to produce effects better fitting their target application. Also our technical analysis in Section 7.1 and our prior discussion showed that *Foley Interpreter* appeared to be able to translate haptic descriptions effectively into audio descriptions most of the time that are closer to the training data set of our employed TTA model. A possible explanation therefore could be that the effects generated based on the audio descriptions from the *Foley Interpreter* form richer audio samples that carry key characteristics meaningful for the haptic experience. As a consequence, when detecting more diverse features in richer audio samples, the audio similarity metric scores lower. As noted before, currently our employed similarity metric was mainly tuned to audible sound effect features between 0-8000Hz, and not to haptic features up to 1khz which were more dominant in the samples used in our work. We highlight that developing more effective similarity metrics that could measure the similarity and quality of haptic effects by detecting haptic features, could not only help increase robustness, but also significantly help increase the overall quality of generated haptic effects when integrating them into feedback loops.

8.4 Top-Down vs. Bottom-Up Haptic Effect Generation Approaches

We classify our approach, the Prompt-to-Touch pipeline, as a Top-down approach to generate haptic effects based on their textual descriptions. It produces audio effects tuned to work as haptic effects. We argue that this approach especially allows us to adapt domain knowledge from the audio experience space to the haptic experience space, therefore not requiring an extensive haptic effect dataset to generalize the approach to different application scenarios. While we show that our Top-down approach yields first promising results, we recognize that other approaches may be more effective. Other work has previously highlighted that the space of haptic experiences may be addressable through a set of basic building blocks such as [97, 112, 113]. Rosenkranz et al. pioneered first steps towards a Bottom-up approach of generating haptic effects based on specified effect profiles and demonstrated its effectiveness for truck driving simulation [96]. We see a text-based bottom up approach and a comparison between both as an interesting and important avenue to explore in the future.

9 Limitations & Future Work

In this work, we demonstrated a proof-of-concept of our proposed pipeline to enhance the haptic experience in different application scenarios by generating dynamic haptic effects from their textual description. However, our results also still indicate significant room for improvement. We discuss several key limitations and possible future investigations in the following that we did not touch on in our discussion.

Methodological Limitations. Within our user studies, we aimed to keep the setup as consistent as possible, to allow evaluating our pipeline and its components across a variety of application scenarios. While we tried to craft convincing application scenarios that establish context for the haptic effects played to the participants, we did not evaluate our pipeline in real application contexts. This specifically concerns the application areas of Immersive XR Experiences, Music & Art, Training & Education, and Industry. While we played vibrations at the hand of the user as common location to evaluate haptic effects, we recognize that for many applications rendering the feedback at different body locations may be more appropriate. Depending on the body location the physiological characteristics may influence the perception and effectiveness of the rendered haptic effects. In addition, we highlight the length of our first user study which could lead to sensory saturation as indicated by two of our participants. Finally, we recognize the limited size of our evaluation set and sample size of our our proof-of-concept investigations. Our method aimed to elicit a first limited set of test scenarios that could represent different application areas which emerged from our focus group. Studies with an extended set of haptic scenarios could provide deeper insights into the true generalizability of our approach in the future.

Spatial Patterns & Feedback Modalities. In this work we focused on the vibrotactile feedback modality. Our haptic feedback device was furthermore limited to one actuator as a first logical step to explore the Prompt-to-Touch pipeline. However, haptic experiences are known to have many facets including sensations with spatial patterns as well as different sub feedback modalities. These can include vibration, stretch, pressure, temperature, and kinaesthetic feedback. For example during our user studies one participant (P19) mentioned they would expect colder sensations for raindrops. This provides insights in how haptic effects generated using our pipeline combined with other modalities could improve their expressiveness in the future. In addition, other device setups including setups beyond a single actuator could be explored in the future to enable spatial patterns.

Temporal Mapping of Generated Effects. A large limitation of our proposed pipeline currently is that the haptic effects cannot be generated in real-time. In addition, some haptic effects are closely connected with audio visual features in complementary audio-video content. During the discussions in our focus group and our elicitation study, participants multiple times suggested scenarios and haptic effect descriptions that relied on a temporal mapping to audio or visual features. In the future it would be therefore interesting to investigate how our pipeline could potentially be used in coherence with prior work that focused on generating real-time haptic feedback based on salient features in audio and video [53, 141].

Foley Interpreter Extensions and Pipeline Expressiveness. While our investigations indicate that the Foley Interpreter component was able to increase the effectiveness of generated haptic effects there is still large room for improvement as current LLMs still lack complete understanding of touch. Custom models, finetuning or improved prompt engineering could increase the effectiveness of our Foley Interpreter component further in the future. For example, our second-step prompt instructs the LLM to *"highlight the key characteristic of the haptic description"* (i.e., the spectral and intensity patterns that evolve over time and define the perceptual quality of the expected audio-haptic experience) and to *"only focus on the most important step part that is most characteristic for the sound effect"*; but the precise wording, level of specificity, and constraints in such prompts (e.g., caption length, exclusion of background noise) are not systematically optimized and could be refined through prompt engineering. Another challenge of our pipeline yet comprises the creation of haptic effects based on descriptions that mention multiple events. Our *Foley Interpreter* design therefore aims to direct the focus on the key characteristic. Allowing the input to define sequences of events could unlock more expressive generation of effects.

In addition, the focus of this work was on the ability to create distinct effects specific to scenarios but did not focus on detailed aspects in the description. We see an exploration that investigates whether attributes used in the effect descriptions can form effective handles to make fine adjustments to semantic characteristics, intensity, or affective aspects as proposed in prior work [45**?**]. In context of our discussion, we highlight that manipulating the intensity through text attributes could form an important aspect to enable gentle effects such as the cats fur in Scenario #01 to be rendered more effectively.

Opportunities for Haptic Effect Design Tools and Automation Processes. This work focused on a first building block for automated text-driven haptic feedback generation. We see potential for several investigations and integrations into different applications in the future. One interesting avenue to explore could be to investigate how the Prompt-to-Touch pipeline can enhance design tools, possibly changing the practices of haptic experts and non-experts. Instead of crafting haptic patterns entirely manually, the haptic designers could provide a description of the haptic experience and our pipeline then auto-generates a corresponding haptic effect. Similar to generative AI audio and image design tools, the application could provide a set of generated examples that inspire the designer, or that the designer can edit and remix. In this context it would be particularly meaningful to compare new haptic design tools based on text-driven feedback generation with traditional haptic feedback design editors. Furthermore, in the era of generative AI and multi-modal LLMs capable of creating wrappers that facilitate easy interfacing between various applications, haptic generation pipelines could complement automated content generation processes similar to emerging trends in the video and audio domains and complement story telling by enhancing existing audio-video content. Here instead of the haptic designer, a multimodal LLM could automatically analyse portions of the audio-video content and describe a potential meaningful haptic experience which is then translated in a haptic effect using our pipeline. By combining prior approaches that detect salient effects in the original audio-video content as well as the generated haptic effect, the haptic effect could be synchronized more appropriately to the other content. We recognize that synchronizing a rhythmic pattern between haptic and audio-video content could constitute a major challenge of our method. One possible way to address this challenge could entail scaling the haptic effect along the temporal dimension until the frequencies of the rhythmic patterns match with each other. We believe next to these examples there could be many other application scenarios analogue to other media design and automation pipelines.

Limited Understanding of Touch in AI Models. As highlighted in our evaluation, large language and text-to-audio models still lack a deep understanding of touch and are biased through their training data which yet entails limited examples of hatpic labels. Also prior work investigating vision large language models (vLLMs) reveal a limited ability in describing tactile features of texture images [28, 136, 144]. This motivates more work investigating how to train multimodal large language models that understand better. A foundation for this is the creation of truly multimodal datasets that label haptic properties in diverse media content. During our investigation we see several approaches in this direction being made. E.g. *HapticLLaMA* aims to produce descriptions based on haptic signals complementary and *HapticCap* contributes a first dataset containing 92,070 haptic-text pairs [40, 41]. We believe more and bigger datasets like these in conjunction with new benchmarking methods will be able to significantly increase the performance of haptic feedback generation pipelines like Prompt-to-Touch or similar approaches.

Ethical Considerations for AI Generated Haptic Effects. Prompt-to-Touch suits to be integrated in different applications including design tools for haptic designers and extended automation pipelines. Moving towards these AI supported design tools and fully autotomized AI driven haptic feedback generation systems brings several ethical implications with it. Particularly because our pipeline integrates two components (an LLM for the Foley Interpreter, and an audio generation model), our pipeline is susceptible to biases in the training data of the used models. This could lead not only to biased interpretations of the haptic descriptions, but also in the generation of the final effect. Several works have highlighted potential ethnicity and gender related differences in the perception of touch experiences [54, 104], indicating that if those populations are not balanced in the training set they may encounter a different haptic experience. Specifically in context of social communication there may be different cultural expectation and acceptance of social touch [104]. In a worst case scenario, models could be exploited in a malicious way to potentially induce physical discomfort when watching a manipulated online video for which a haptic effect is automatically generated. Next to the generated haptic effect pattern, the impact of the effect can largely depend on the location, configuration and type of the used actuation system. Finally, while increasing effort is put to prevent them, machine learning models are known to hallucinate which could result in unwanted generations of haptic effects. While further work is required to investigate ethical implications and potential issues, we see three approaches to mitigate potential negative effects: 1) To take ethical considerations of touch into account and ensure balancing the training and evaluation set based on populations that likely interpret or experience touch differently, or are likely underrepresented. 2) To implement a filtering mechanism that prevents potential harmful haptic effects at different stages of the effect generation process. 3) To limit characteristics of the physical haptic actuator and avoid body locations that could cause physical or mental discomfort.

10 Conclusion

In this work we introduced *Prompt-to-Touch*, a multi-step pipeline to generate haptic effects for different applications based on a textual description of the haptic experience. We proposed four core components to convert the textual input into a haptic effect, including a *Foley-Interpreter*, *Text-to-Audio Generator*, *Dynamic Audio Processor*, and *Post-Processor*. We first conducted a focus group to collect a test application set and elicited haptic effect descriptions from non experts. We then conducted two human evaluation studies and a technical analysis to evaluate our pipeline for its ability to produce meaningful haptic effects. Results showed that effects generated using our pipeline tend to reflect the haptic effect descriptions and enhance haptic experiences across different application scenarios. We reported several insights on the performance of different pipeline components that allowed us to identify limitations of our pipeline and suggest several starting points for future improvements. We hope that our pipeline inspires future investigations that explore its integration into more accessible design tools and automation processes for haptic feedback generation.

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A Appendix: Foley Interpreter Instructions

Step 1, System Message:

"*You are an expert haptic feedback designer.*"

Step 1, Instruction:

"*Your job is to come up with a description of an audio effect for the description of the haptic touch experience. That means you describe an audio effect that is likely to exist that could resemble the key characteristics of the described haptic touch experience. If the description contains non audible aspects, try imagine what audio effects may result in the haptic experience. Here is the description of the haptic touch experience: '[HAPTIC EFFECT DESCRIPTION]'. How would you describe the translated audio effect? Think step by step. If applicable and meaningful to the sound, describe how the sound effect evolves step by step. Keep in mind that only one speaker is available to play the effect.*"

Step 2, System Message:

"*You are an expert in writing audio captions for sound effects.*"

Step 2, Instruction:

"*Now output the final audio effect description in the format of a short audio caption. The effect should be unique and highlight the key characteristic of the haptic description. If the sound effect evolves over time, only focus on the most important step part that is most characteristic for the sound effect. Ensure that the caption is in double-quotes. The audio caption should be short, and descriptive, focusing on non-technical descriptions of a sound effect. Ignore background noise and avoid humming.*"

B Application Scenarios

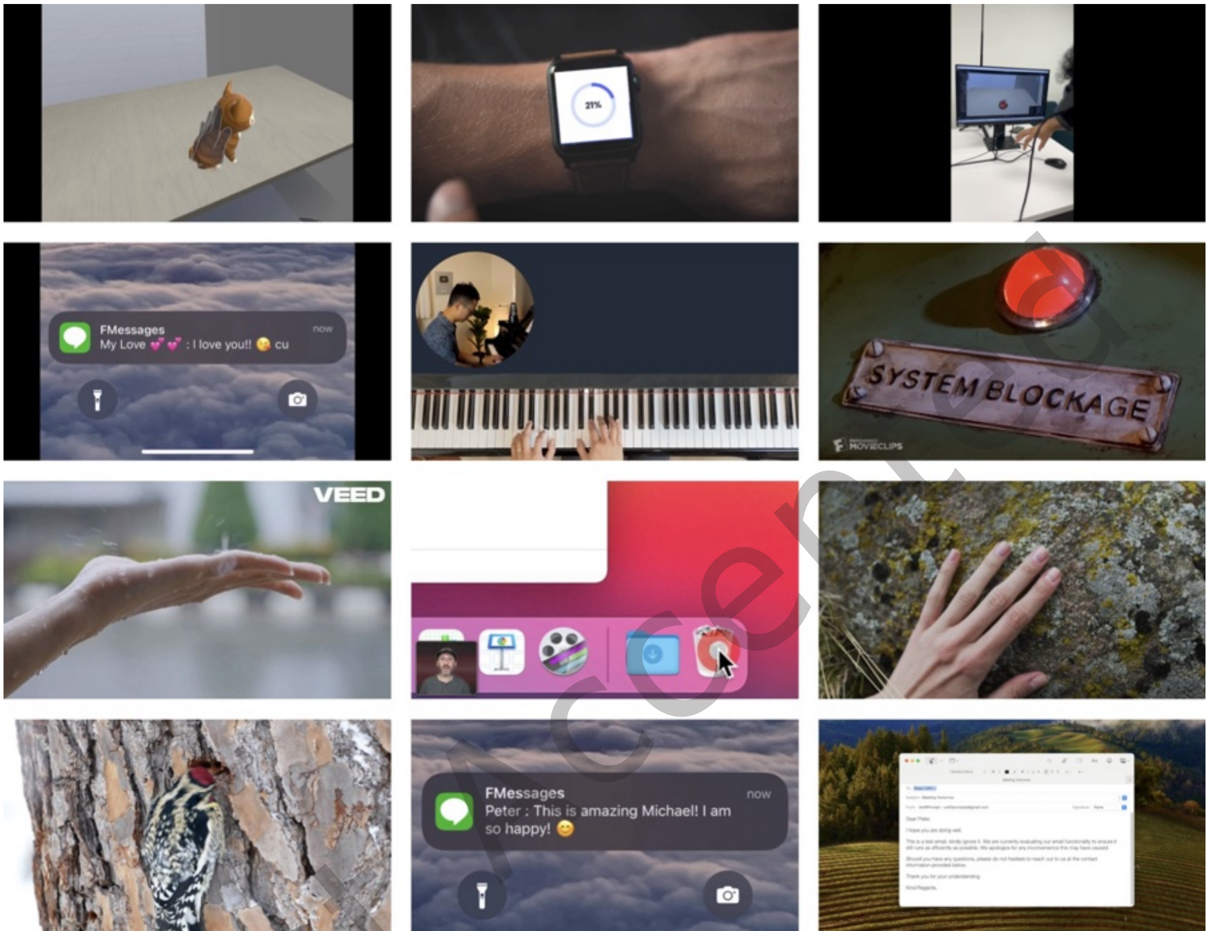

Fig. 14. **Application Scenario Test Set Visuals:** Visuals of all 12 application scenarios as described in Table 1 ordered from top-left to bottom-right.

C Additional Cross-validation Scenarios

#	Application Area	Scenario Description	Haptic Effect Description	Foley Phrase
1	Medical	”Imagine you are palpating a patient’s arm to check for swelling or tenderness.”	”Fingers gently pressing and gliding over soft tissue, detecting subtle variations in firmness and texture.”	”A gentle rustle morphs into a soft, crinkling touch, capturing the delicate press and glide over varied textures.”
2	Medical	”Imagine inserting a needle for a precise injection.”	”A subtle, brief resistance giving way instantly.”	”A gentle snap followed by a soft, quick pop, capturing the essence of tension seamlessly giving way.”
3	Sports	”Imagine running barefoot over a gravel trail.”	”Rapid, uneven pressure under the soles, alternating between sharp pokes from stones and softer contact with earth.”	”A sequence of sharp, crisp clicks evoking stones underfoot transitions into a soft, resonant thud, mimicking the sensation of stepping onto yielding earth, creating an alternating rhythm of hard and soft impacts.”
4	Sports	”Imagine swinging a baseball bat to hit a fast pitch.”	”A sharp, brief vibration traveling through the hands.”	”A sharp, snapping sound with an immediate flickering shift, like a quick whip crack echoing ever so briefly.”
5	Accessibility	”Imagine reading Braille on a tactile page with your fingertips.”	”Light, rhythmic touches moving across raised dots, feeling distinct bumps.”	”Soft, rhythmic tapping followed by occasional crisp clinks, creating a pattern akin to gentle fingers discovering textured dots on a surface.”
6	Accessibility	”Imagine feeling the vibration cues from a smartphone as an accessibility aid.”	”Short, patterned vibrations under fingertips, varying in intensity and rhythm.”	”A series of crisp, rhythmic clicks with varying loudness, resembling a ticking clock interspersed with moments of silence, mimicking the pattern of vibrating fingertips.”

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